MAPPING OF ALIENABLE LAND IN BHUTAN

NATIONAL LAND COMMISSION

Technical Report 1: Baseline Assessment Using GIS-based Multi-Criteria Decision Analaysis (MCDA)

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National Land Commission Secretariat

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Mapping of Alienable Land in Bhutan
Technical Report 1: Baseline Assessment Using GIS-Based Multi-Criteria Decision Analysis (MCDA)

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FOREWORD

Bhutan's limited land resources are increasingly subject to competing demands arising from development, conservation, infrastructure expansion, and emerging economic initiatives. These challenges necessitate more spatially explicit, evidence-based, and integrated approaches to land use planning and decision-making. Strategic initiatives, including the Integrated Land Use and Spatial Planning Framework (ILUSP), the National Land Use Zoning (NLUZ) initiative, the Comprehensive National Development Plan (CNDP), sectoral master plans, and the 10X Economic Vision, collectively underscore the importance of strengthening geospatial and analytical capabilities to support coordinated and sustainable development.

Against this backdrop, the Mapping of Alienable Land in Bhutan has been undertaken as a preliminary technical assessment to identify potential alienable land through the application of Geographic Information Systems (GIS)-based Multi-Criteria Decision Analysis (MCDA).

This report presents the methodological foundation for assessing alienable land potential across the country. It provides a systematic, transparent, and replicable approach for identifying areas that may be suitable for future development while taking into account key constraints and regulatory requirements. The methodology is designed to remain adaptive and can be further strengthened through the incorporation of additional datasets, planning parameters, and emerging information as they become available. The current outputs have been refined through desktop verification and field validation, as detailed in Technical Report 2: Desktop Verification and Field Validation.

The outputs of this study are expected to support the identification and management of available State land for economic and developmental purposes and contribute to the ongoing development of the Land Bank Management System (LBMS). They are also expected to guide sectoral planning, master planning exercises, and other strategic national development initiatives, including the Gelephu Mindfulness City. The National Land Commission Secretariat remains committed to advancing data-driven decision-making and promoting the effective use of geospatial technologies in support of sound land governance and the long-term development aspirations of the Kingdom of Bhutan, guided by the noble vision of His Majesty the King.

(Tshering Gyaltshen Penjor)
Secretary
National Land Commission Secretariat

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The National Land Commission Secretariat (NLCS) gratefully acknowledges WWF Bhutan for their generous support as the primary sponsor of this initiative. Their contribution, including support for capacity development at the Asian Institute of Technology (AIT), Thailand, played a vital role in strengthening the technical capacity required for this study.

We also extend our sincere appreciation to the Royal Government of Bhutan for its continued support in promoting integrated land use planning and sustainable development initiatives.

Special recognition is extended to all contributing agencies under the Centre for Geo-Information and the Bhutan National Spatial Data Infrastructure (NSDI) system for sharing essential datasets used in this study. These include the Department of Forests and Park Services (DoFPS), Department of Human Settlement (DHS), Department of Surface Transport (DoST), and the National Land Commission Secretariat (NLCS).

The Geo-Informatics Division (GID), NLCS, acknowledges the dedication and technical efforts of the project team in data processing, spatial analysis, model development, and cartographic production, which made this report possible.

EXECUTIVE SUMMARY

Context

Bhutan has limited developable land due to its rugged topography, with only a small portion of the country suitable for development. Within this context, identifying **Alienable Land** as state-owned land potentially suitable for allocation for multiple development uses is critical for sustainable spatial planning.

> **Definition: Alienable Land** refers to state-owned land identified through spatial analysis as potentially suitable for allocation for various purposes, including agriculture, settlements, industrial activities, and infrastructure development. This classification is indicative in nature and is subject to confirmation through legal review, environmental assessment, and field verification.

This report establishes the baseline for mapping of alienable land in Bhutan using a GIS-based Multi-Criteria Decision Analysis (MCDA).

Modeling Scenarios (The Land Funnel)

To provide a transparent analysis, the study progressively constrained the national land mass through three analytical scenarios:

- **Option 1 (Theoretical Baseline):** In this analysis scenario, no existing constraints were applied. This generates the gross suitable potential areas of alienable land including river beds, existing towns and cities due to flat slopes and favorable topography.
- **Option 2 (Physical and Legal Reality):** In this analysis scenario, the following constraints were applied. Existing cadastral plots (private land), specific LULC limitations (landslides, moraines, rock outcrops, water bodies, glaciers), rivers, and roads were applied as constraints.
- **Option 3 (Conservation Constraints):** This option provides the Final Net Baseline output of the suitable Alienable Land in Bhutan. For this, in addition to the constraints applied in option 2, non-negotiable constraints for protected areas (Core, Transition, and Biological Corridors) and Ramsar-designated wetlands were applied, ensuring total alignment with environmental conservation mandates.

Key Findings

The assessment of Alienable Land suitability under three scenarios showed a significant reduction in alienable land. As the analysis moved from the unconstrained scenario (Option 1) to the fully constrained operational scenario (Option 3), the area identified as “Highly Suitable” (S1) decreased by 58.88% (from 1,528,207.01 acres to 628,419.40 acres). This reduction was mainly due to steep terrain, legal restrictions, and environmental conservation policies.

In contrast, Moderately Suitable (S2) land showed strong stability, with about 2,431,447.40 acres remaining available. This category forms the main land reserve for future development in the country.

At the regional level, there is a clear difference whereby southern and central-western Dzongkhags such as Wangdue Phodrang, Samtse, and Dagana have higher development potential, while northern and central-eastern Dzongkhags are more limited due to strict conservation requirements.

A summary table (Table ES.1) and comparative chart (Figure ES.1) presenting the national-level distribution of suitability classes across the three scenario-based land funnel options are provided on the following page.

Use of This Report

This report serves as a macro-level decision-support tool for the pre-screening and strategic planning of alienable land. The results are indicative in nature and require site-specific validation prior to final land allocation.

Technical Report 2

Highly suitable areas identified in this study were subjected to systematic desktop verification, with selected sites further validated through field-based assessment, as detailed in Technical Report 2.

Overview of National Suitability Results

Suitability Class	Option 1 (Gross Potential)	Option 3 (Final Net Baseline)	Percentage Reduction
Highly Suitable (S1)	1528207.01	628419.40	58.88%
Moderately Suitable (S2)	4062288.58	2431447.40	40.15%
TOTAL USABLE BANK	5590495.59	3059866.80	45.27%

Table ES.1: Scenario Comparison: Option 1 vs Option 3

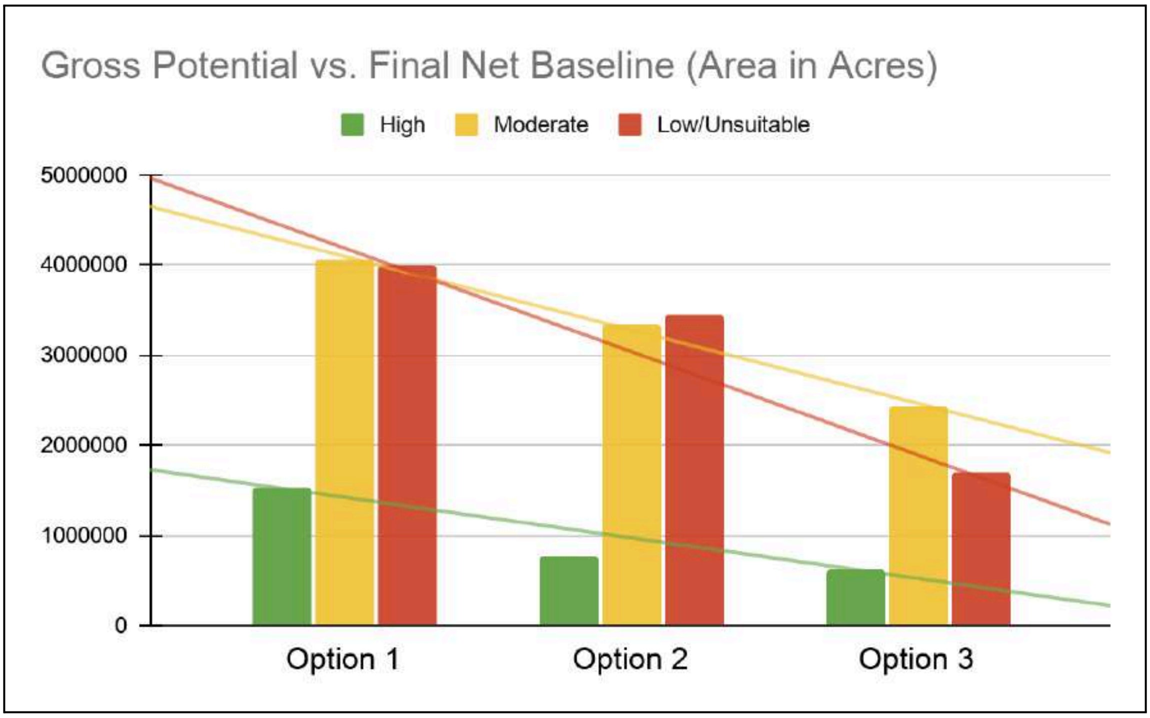

Figure ES.1: Comparison of National Alienable Land Inventory through the Land Funnel

ACRONYMS AND ABBREVIATIONS

Acronym	Full Form
AHP	Analytic Hierarchy Process
CGI	Centre for Geo-Information
CR	Consistency Ratio
CNDP	Comprehensive National Development Plan
DEM	Digital Elevation Model
DoFPS	Department of Forests and Park Services
DoST	Department of Surface Transport
FMU	Forest Management Unit
GIS	Geographic Information System
GID	Geo-Informatics Division
ILUSP	Integrated Land Use and Spatial Planning
LBMS	Land Bank Management System
LULC	Land Use and Land Cover
MCDA	Multi-Criteria Decision Analysis
MCE	Multi-Criteria Evaluation
MMU	Minimum Mapping Unit
NLCS	National Land Commission Secretariat
NLUZ	National Land Use Zoning
NSDI	National Spatial Data Infrastructure
NSSC	National Soil Services Centre
SAR	Synthetic Aperture Radar
TCC	Tree Canopy Cover
TR	Thinness Ratio
UAV	Unmanned Aerial Vehicle
WLC	Weighted Linear Combination

LIST OF FIGURES

LIST OF TABLES

TABLE OF CONTENTS

1. BACKGROUND AND SIGNIFICANCE

1.1 Context

Bhutan faces escalating pressure on land resources. Rapid socio-economic development, population growth, and strict environmental conservation commitments require a highly scientific approach to land use. The **Integrated Land Use and Spatial Planning (ILUSP)** and **National Land Use Zoning (NLUZ)** initiatives were introduced to shift from historically reactive land allocation to proactive, evidence-based spatial planning.

1.2 Significance

- **Primary Input:** Serves as the foundational spatial dataset for the Land Bank Management System (LBMS).
- **Informed Decision-Making:** Empowers state land management and strategic allocation by identifying viable land parcels.
- **Strategic Balance:** Reconciles high-growth economic development goals with strict environmental stewardship.
- **National Baseline:** Provides a unified and reliable reference framework for multi-sector national investment, integrated land use planning, and sustainable land management across the country.

2. OBJECTIVES OF THE STUDY

2.1 Main Objective

- To identify and map alienable state land using GIS and Remote Sensing-based Multi-Criteria Decision Analysis (MCDA).

2.2 Specific Objectives

- **Framework Development:** Develop a standardized GIS–MCDA framework for Alienable Land Mapping (ALM).
- **Database Creation:** Create a national spatial database of Potential Alienable Land (PAL).
- **Suitability Assessment:** Assess land suitability under physical, legal, and environmental constraints.
- **Decision Support:** Support spatially informed decision making.

3. SCOPE AND EXPECTED BENEFITS

3.1 Scope

- **Coverage:** Comprehensive coverage of all 20 Dzongkhags of Bhutan.
- **Theme:** General Suitability (Macro-level Alienable Land identification).
- **Report Level:** Technical Report 1 (National-level baseline).
- **Indicative Nature:** Results are macro-level; site-specific allocation requires field validation.

3.2 Expected Benefits

- **Evidence-Based Allocation:** Supports informed and transparent land allocation decisions.
- **Development Prioritization:** Identifies priority areas for economic development.
- **Data Integration:** Strengthens NSDI and cross-sectoral spatial data integration.
- **Sustainable Planning:** Enables balanced and sustainable land use planning.

4. METHODOLOGY (GIS–MCDA FRAMEWORK)

4.1 Overall Approach

The study applies a GIS based Multi-Criteria Decision Analysis (MCDA) framework implemented through Multi-Criteria Evaluation (MCE), using the Analytic Hierarchy Process (AHP) for weighting and the Weighted Linear Combination (WLC) method for suitability aggregation. Spatial data preparation and post-processing were conducted in QGIS and ArcGIS Pro, while suitability modelling was performed in the MCE module of TerrSet liberaGIS. The overall workflow from data preparation to suitability modelling and land funnel scenario analysis is shown in Figure 4.1.

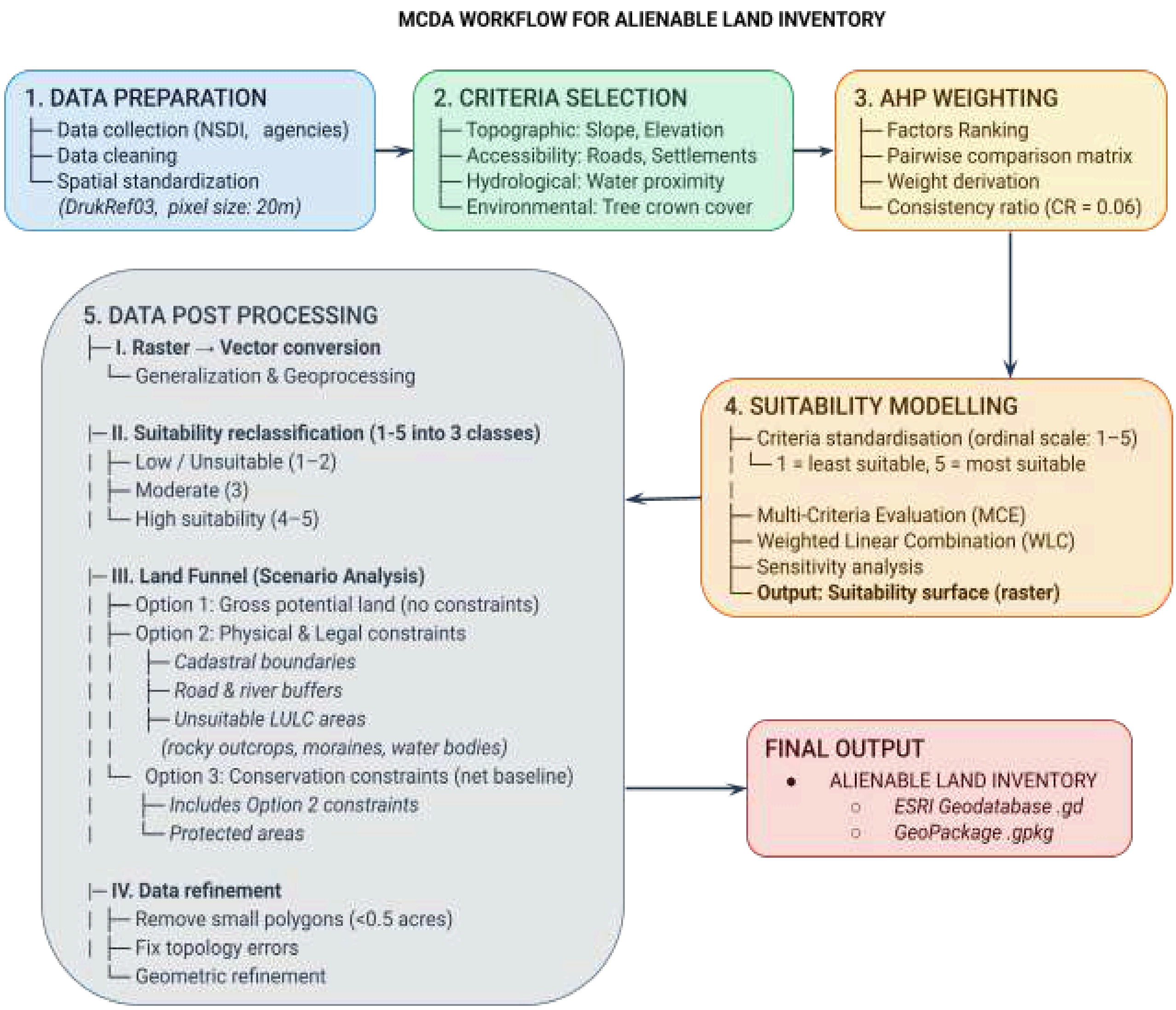

Figure 4.1.1: Multi-Criteria Decision Analysis (MCDA) Workflow for Alienable Land Suitability Assessment Using AHP and GIS-Based Spatial Modelling

4.2 Key Steps

Step 1: Data Preparation

- **Data Acquisition:** Spatial datasets were collected from relevant national agencies and Bhutan NSDI System.
- **Standardization:** All datasets were standardized to a 20 m spatial resolution and projected to the DrukRef03 coordinate system (EPSG: 5266) to ensure consistent multi-source raster integration and alignment.

Step 2: Criteria Selection

- **Topographic Factors:** Slope and elevation.
- **Accessibility Factors:** Proximity to roads and settlements.
- **Hydrological Factor:** Proximity to water bodies.
- **Environmental Factor:** Tree crown cover.

Criteria were selected based on study objectives and the availability of standardized spatial datasets.

Step 3: Weighting (AHP)

- **Method:** The Analytic Hierarchy Process (AHP) was applied to assign weights. The pairwise comparison matrix was developed by the Technical Working Group (TWG) based on expert judgment.
- **Key Drivers:** Slope (0.48) and proximity to settlements (0.21) emerged as the dominant factors influencing suitability.

The relative importance of criteria was determined via a Pairwise Comparison Matrix. The factor ranking and final weights for Suitability Mapping are as follows:

Rank	Factor	Weight	Justification of AHP Weights
1	Slope	0.485	Highest weight as the primary physical constraint in Bhutan's rugged terrain, directly controlling land stability and construction feasibility
2	Proximity to Settlements	0.213	High weight due to concentration of development around existing settlements with established infrastructure and high expansion feasibility
3	Proximity to Water	0.136	Moderate weight as water is essential for domestic and agricultural needs, but largely aligns with existing settlement patterns
4	Proximity to Roads	0.093	Lower weight as accessibility is important but road networks generally follow settlement distribution and terrain constraints
5	Tree Canopy Cover (TCC)	0.043	TCC is given a low weight to act as a secondary environmental screening factor, ensuring it does not dominate the suitability model while still discouraging inclusion of densely forested areas.
6	Elevation	0.031	Lowest weight as its influence is indirect and largely captured through slope and broader terrain conditions
	Total	**1.00**	

Table 4.2.1: AHP-Derived Weights and Justification of Factors Used for Suitability Mapping

Technical Note: *The model achieved a Consistency Ratio (CR) of 0.0347 for a 6 × 6 pairwise comparison matrix (n = 6). Since the CR is below the 0.10 threshold, the weighting is considered mathematically consistent and logically sound. The detailed pairwise comparison matrix and AHP computation steps are provided in Appendix-I, Section B.*

Step 4: Suitability Modeling and Surface Generation

- **Methodology:** Suitability modelling was performed using the Weighted Linear Combination (WLC) approach within the Multi-Criteria Evaluation (MCE) module of TerrSet liberaGIS Version 20.0.1.
- **Standardization:** All raster input datasets were standardized to a common ordinal scale ranging from 1 to 5, where 1 represents low or unsuitable conditions and 5 represents highly suitable conditions. This preprocessing step was carried out prior to suitability modelling to ensure comparability among all criteria layers.
- **Aggregation:** Application of AHP-derived weights to factor layers to generate a continuous suitability surface at a 20 meter spatial resolution.

Detailed reclassification thresholds and the WLC formulation are provided in Appendix-I, Sections A and C.

Step 5: Option Analysis (Scenario Modeling)

This stage integrates scenario-based land filtering and geometric refinement to convert theoretical suitability outputs into a practical **Alienable Land Inventory**.

- **Scenario Analysis (Land Funnel Approach):**
  This step progressively refines land suitability through a filtering sequence that moves from broad physical potential to legally and environmentally constrained land.
  - **Option 1: Gross Potential**
    Identification of maximum suitability based only on physical terrain conditions, without any restrictions.
  - **Option 2: Physical & Legal Constraints**
    Exclusion of physically and legally unsuitable areas by applying LULC restrictions (e.g., landslides, rocky outcrops, snow and glaciers, moraines, and water bodies), along with slope constraints (>35°), river and road buffers, and cadastral land records.
  - **Option 3: Final Net Baseline**
    Further refinement of Option 2 by applying the Conservation Mask to exclude Ramsar sites and protected areas (core zone, transition zone, and biological corridors) to derive the operational land baseline.
- **Post-Processing (Geometric Refinement):**
  This step ensures the spatial outputs are clean, usable, and suitable for inventory generation.
  - Conversion of final suitability raster into vector polygons.
  - Removal of polygons smaller than 0.5 acres (Minimum Mapping Unit).
  - Shape optimization using Thinness Ratio (TR) to eliminate sliver and irregular polygons

National contraction statistics comparing Options 1 and 3 are detailed in Section 5.2. For a full breakdown, Appendix-II provides Dzongkhag-wise comparative tables and maps, while Appendix-I, Section D contains the mathematical formulations for the Thinness Ratio (TR) and cleaning thresholds.

5. RESULTS AND ANALYSIS

5.1 Presentation of Results and Cartographic Standards

Presentation of Results

The results of the GIS based MCDA analysis are presented through a combination of spatial maps, statistical tables, and graphical visualizations. Maps are used as the primary medium to illustrate the spatial distribution of land suitability across Bhutan under different constraint scenarios. These are supported by quantitative analysis to provide both spatial and numerical insights into alienable land availability. The integration of spatial and statistical outputs enables both visual interpretation and data-driven decision making.

Cartographic Standards and Legend Note

All maps presented in this report follow a standardized cartographic scheme to ensure consistency, readability, and comparability across all outputs.

Suitability Classes:

- Highly Suitable (S1): Represented in Green
- Moderately Suitable (S2): Represented in Yellow
- Low or Unsuitable: Represented in Red

Mapping Standards:
All final outputs are presented in vector format following post-processing and geometric refinement. A unified legend, north arrow, and scale bar are consistently applied across all maps. Dzongkhag and national boundaries are uniformly overlaid to ensure consistent spatial reference. Symbology, classification, and labeling are standardized across all map products to ensure comparability and visual consistency.

5.2 National Suitability Maps

The land suitability is shown through three stages of refinement (land funnel approach):

- **Option 1: Gross Potential Map**
- **Option 2: Physical and Legal Constraints Map**
- **Option 3: Final Net Baseline Map**

These represent progressive refinement of land suitability, from theoretical potential to a more realistic baseline of the **Alienable Land Inventory**, as described in Section 4.2 (Step 5).

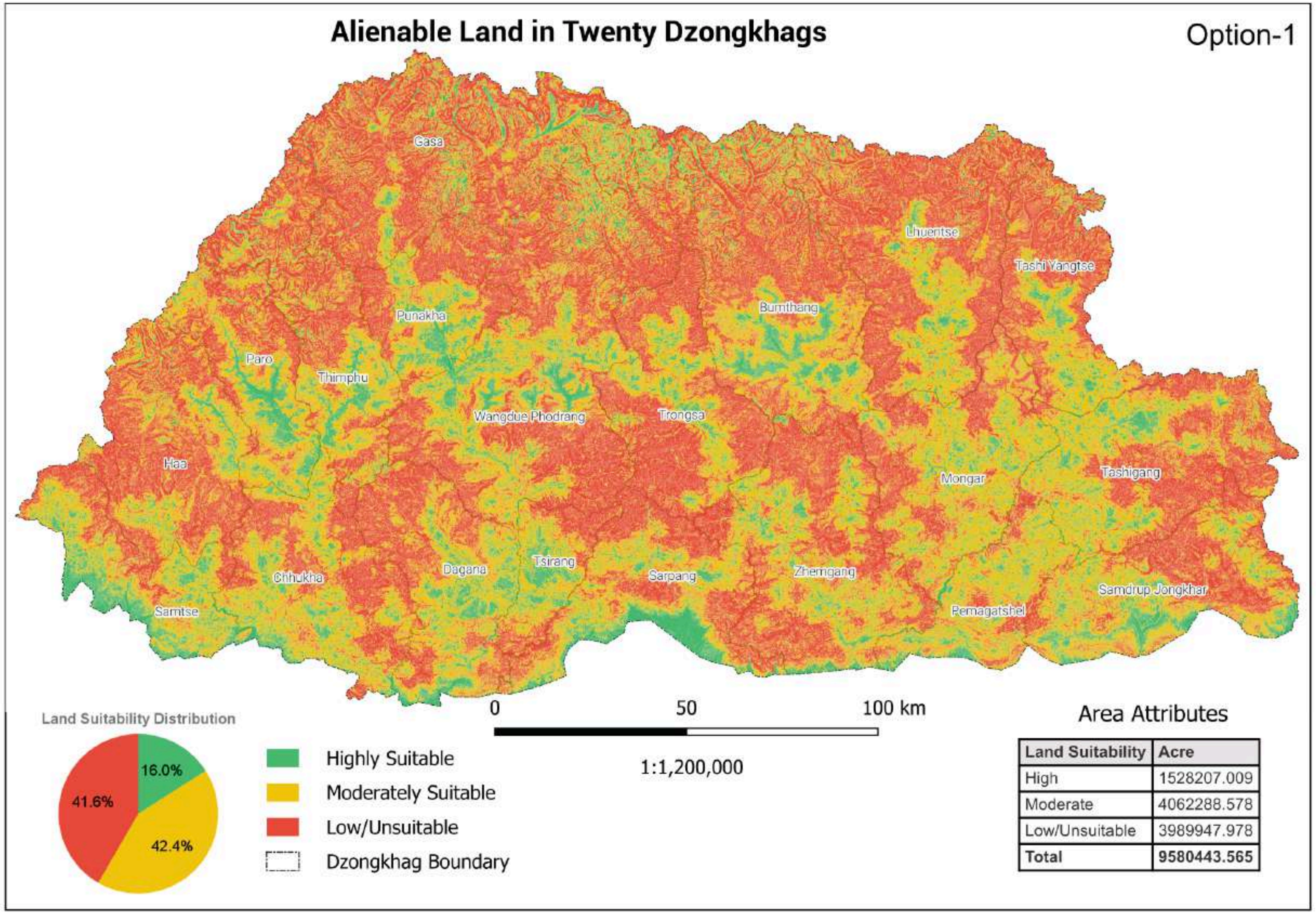

Figure 5.2.1 – National Suitability Map (Option 1: Unconstrained)

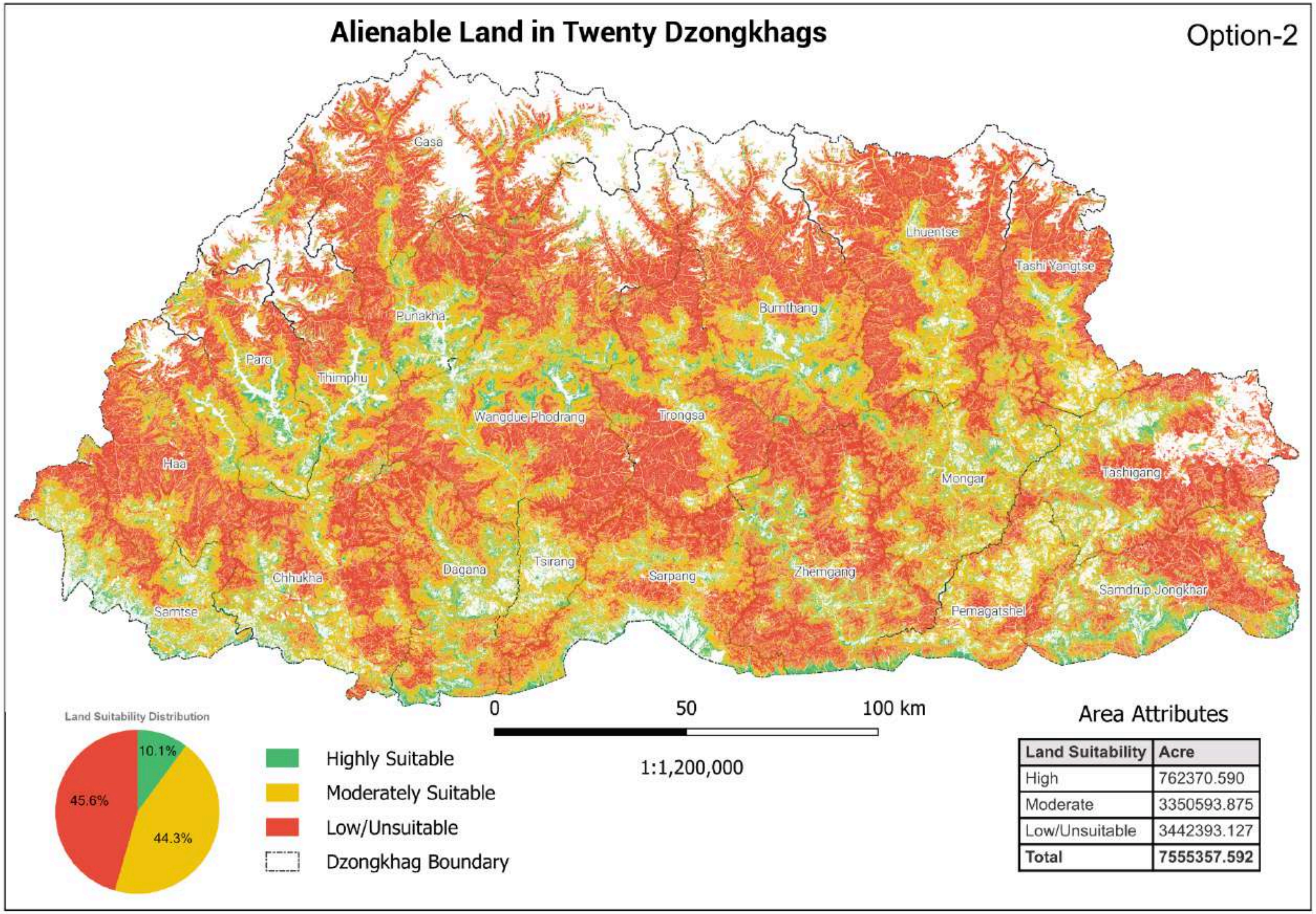

Figure 5.2.2 – National Suitability Map (Option 2: Physical & Legal Constraints)

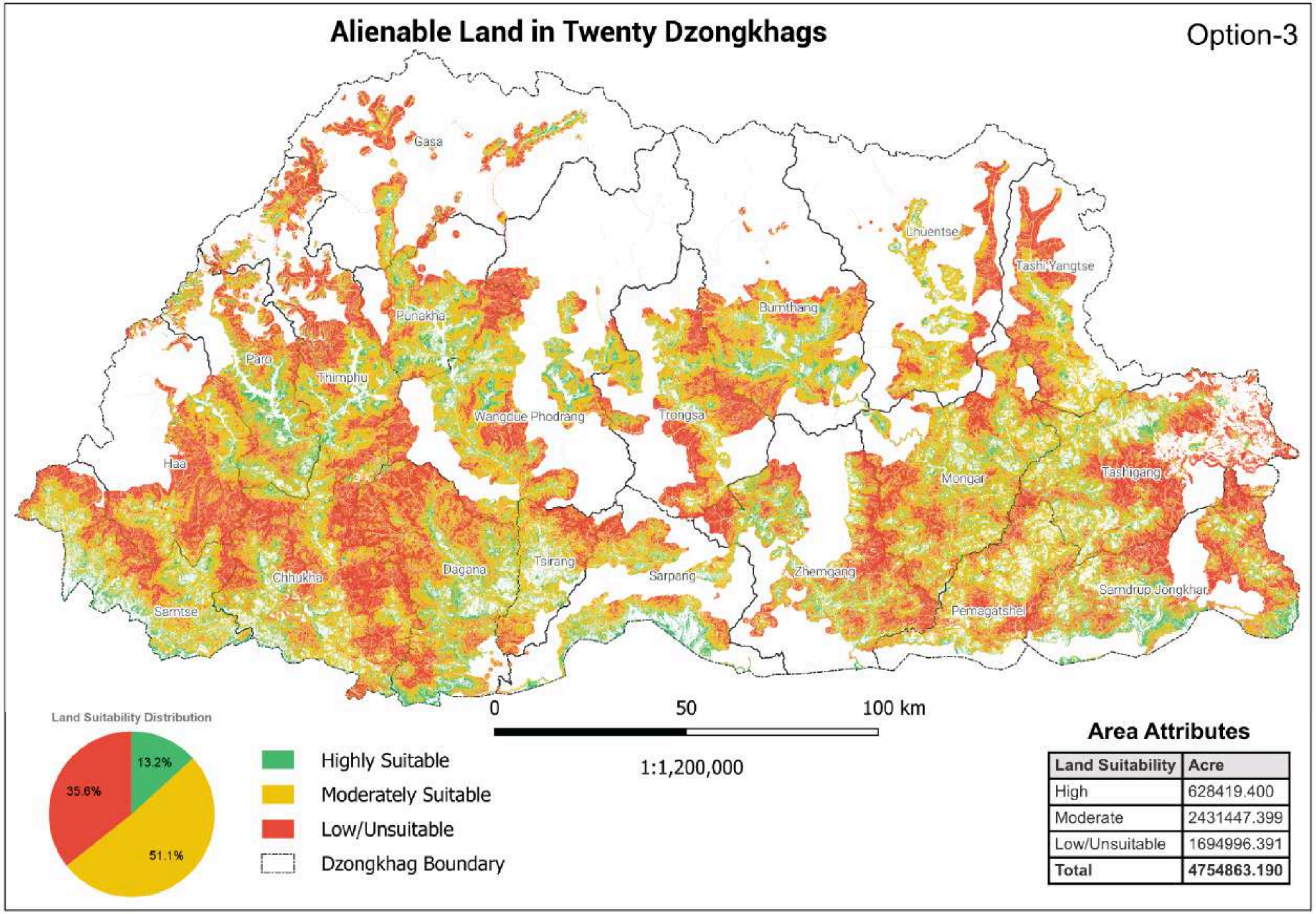

Figure 5.2.3 – National Suitability Map (Option 3: Final Net Baseline after discounting conservation constraints)

5.3 Scenario-wise Results (National Scale Analysis)

From Option 1 to Option 3, all suitability classes exhibit a reduction in available land area. The results show a progressive decline in High, Moderate, and Low/Unsuitable classes under increasing spatial constraints for identifying alienable land. This reflects a systematic decrease in both developable land and conditionally suitable land across all scenarios.

Suitability Class	Option 1	Option 2	Option 3	Change (O1 - O3)	% Reduction
High	1,528,207.01	762,370.59	628,419.40	899,787.61	58.88%
Moderate	4,062,288.58	3,350,593.87	2,431,447.40	1,630,841.18	40.15%
Low/Unsuitable	3,989,947.98	3,442,393.13	1,694,996.39	2,294,951.59	57.52%

Table 5.3.1:Option-wise Suitability Change with Percentage Reduction

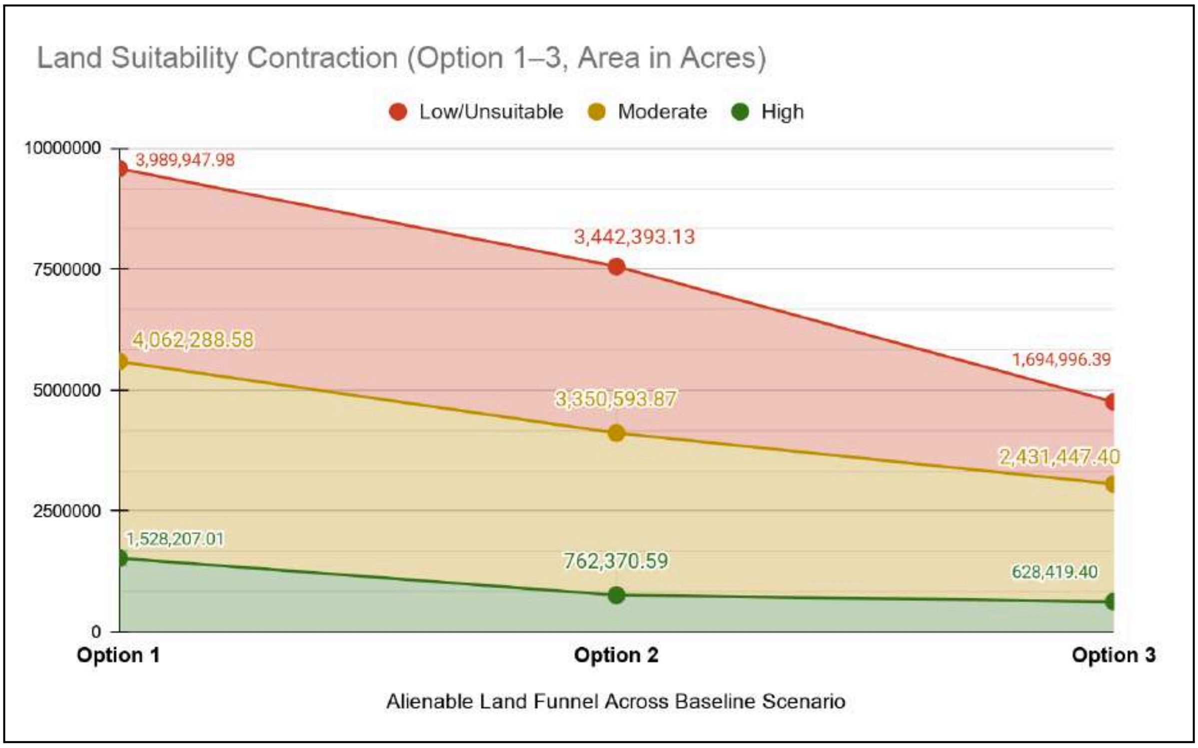

Figure 5.3.1: National Land Suitability Reduction Across Scenario Options

Result Interpretation

At the national scale, the results indicate a progressive contraction of alienable land as additional spatial constraints are introduced through the scenario framework defined in Step 5 of Section 4.2. This reflects the transition from a theoretical suitability surface to a more constrained planning reality.

The most pronounced reduction is observed in High suitability land, highlighting its sensitivity to cumulative constraint application. Moderate and Low/Unsuitable categories also show substantial reductions, indicating that constraint effects are not limited to a single suitability class but operate across the full spatial distribution.

Collectively, this demonstrates that a large proportion of land identified under unconstrained conditions does not remain viable once planning constraints are applied. Option 3 therefore represents the most realistic and policy-relevant baseline for national-level land use planning and future development prioritization.

5.4 Dzongkhag-wise Area Statistics (Final Net Baseline)

The data below quantifies the final, net baseline for **Alienable Land Inventory** (Option 3).

Dzongkhag	High Suitability (S1) Acres	Moderate Suitability (S2) Acres	Low/Unsuitable Acres	S1+S2 (% of Area)
Bumthang	40069.76	131241.06	67562.76	71.72
Chhukha	41322.51	198226.31	156376.62	60.50
Dagana	45282.57	167812.32	138573.52	60.60
Gasa	12317.99	48807.36	54971.61	52.65
Haa	17161.12	106027.26	118170.97	51.04
Lhuentse	16788.80	84213.85	51308.65	66.31
Monggar	37405.99	199087.58	111155.25	68.03
Paro	37681.63	109355.52	55937.59	72.44
Pemagatshel	21091.94	99451.52	43403.87	73.53
Punakha	27801.59	81976.53	44521.47	71.15
Samdrup Jongkhar	43076.89	144404.03	99991.64	65.22
Samtse	48431.94	127855.67	66201.70	72.70
Sarpang	34918.23	74461.92	37429.25	74.50
Tashigang	31405.57	164348.76	135121.49	59.16
Thimphu	33721.43	132366.99	117101.08	58.65
TrashiYangtse	14709.39	80631.96	61833.80	60.66
Trongsa	20755.24	99270.21	92445.02	56.49
Tsirang	15415.51	55400.43	35336.28	66.71
Wangdue Phodrang	58714.42	175375.12	102419.26	69.56
Zhemgang	30346.86	151133.01	105134.56	63.32
National	**628419.40**	**2431447.40**	**1694996.39**	**64.35**

Table 5.4.1: Dzongkhag-wise Land Suitability Distribution under NET Baseline Scenario (Option 3)

5.5 Strategic Dzongkhag Insights: Development vs. Conservation

The analysis reveals a clear spatial contrast between Dzongkhags with strong development potential and those governed by conservation mandates.

High Potential Dzongkhags are characterized by favorable topography, better infrastructure accessibility, and limited overlap with protected areas. Wangdue Phodrang exhibits the highest extent of S1 land nationally, influenced by suitable terrain conditions and its relatively large geographical area among all Dzongkhags. The Southern Industrial Belt comprising Samtse, Dagana, Chhukha and Samdrup Jongkhar forms a continuous spatial corridor of high development potential supporting industrial expansion and trade connectivity.

Conservation and Terrain Constrained Dzongkhags experience substantial reductions in suitability under Option 3 due to the combined influence of protected areas, biological corridors and steep mountainous terrain. Gasa shows near total absence of high suitability land primarily due to the dominance of Jigme Dorji National Park and extreme altitude conditions. Similarly, Zhemgang and Lhuentse remain highly constrained, reflecting both ecological sensitivity and rugged topographic conditions.

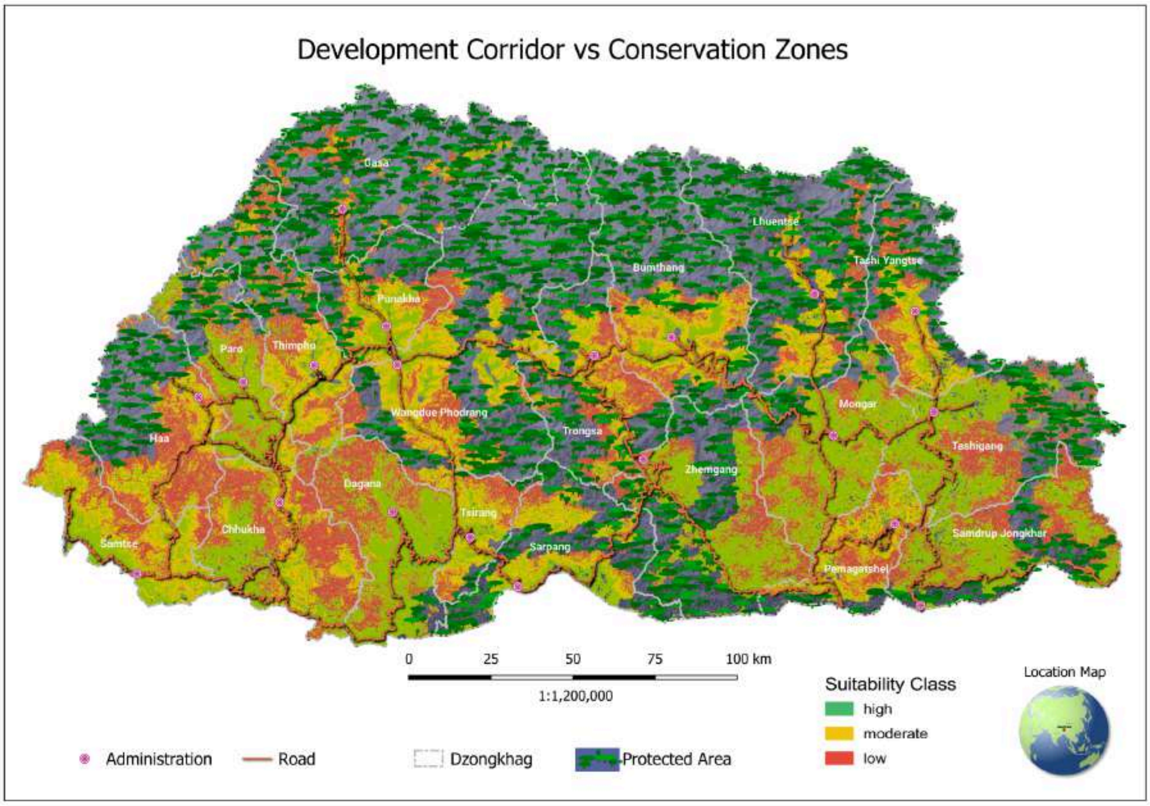

Figure 5.5.1: National Spatial Divide Between Development Corridors and Conservation Zones

5.6 Key Systemic Insights

The results of the MCDA-based suitability analysis reveal three overarching systemic patterns that collectively define the national land suitability structure under progressively applied spatial constraints.

Conservation Trade-off: Environmental protection zones significantly reduce highly suitable land under Option 3, acting as the dominant constraint in the final spatial distribution of alienable land.

Resilience of Moderately Suitable (S2) Land: S2 land remains the most stable category across all scenarios and serves as the primary functional buffer for future land allocation and phased development planning.

Regional Spatial Polarization: A clear spatial divide exists between southern development corridors and northern conservation dominated regions, strongly shaping the pattern of alienable land.

Overall, the MCDA results demonstrate a systematic reduction in land availability as spatial constraints are progressively applied. High suitability land is significantly constrained under the final scenario, while moderately suitable land remains the most stable and strategically important category. The analysis highlights a pronounced spatial polarization between development oriented southern Dzongkhags and conservation constrained northern regions, providing a critical baseline for national land use planning and decision support.

6. MCDA MODEL VALIDATION

6.1 Stability Testing of the MCDA Model

The robustness of the AHP-based MCDA model was evaluated through a ±10% perturbation of criterion weights to assess the stability of spatial suitability outputs. This sensitivity testing examines how variations in input weights influence the resulting spatial distribution of land suitability and identifies the relative influence of each factor.

The analysis was performed by systematically varying individual factor weights while maintaining all other parameters constant. The resulting changes in the spatial output were measured using variance and average cell difference statistics.

In addition, selected sensitivity scenarios were spatially overlaid with the base suitability map. These overlay results confirmed that although localized shifts in suitability classes were observed, the overall spatial structure of land suitability remained stable and consistent across scenarios.

6.2 Variance and Sensitivity Interpretation

Variance represents the degree of spatial change in the suitability output resulting from adjustments in criterion weights. Higher variance indicates greater sensitivity, where small changes in weight produce more significant shifts in spatial allocation.

The results reveal a clear hierarchy of factor influence within the MCDA model. Proximity to settlements exhibits the highest sensitivity (variance = 10,317.40), indicating that accessibility to settlements is the most influential driver of land suitability and a primary determinant of spatial allocation patterns.

Tree Canopy Cover (TCC) shows the second highest sensitivity (8,113.01), reflecting the influence of forest density and ecological constraints. Slope, despite having the highest assigned weight in the model, shows moderate sensitivity (7,746.95), indicating its role as a stable physical constraint governed by Bhutan’s topography.

Proximity to water (6,913.12) and elevation (6,820.25) exhibit comparable moderate sensitivity, suggesting balanced contributions to spatial variation. These factors influence environmental suitability but do not dominate spatial redistribution under weight perturbation.

Proximity to roads shows the lowest sensitivity (4,344.49), indicating comparatively stable influence on the MCDA output. This reduced sensitivity is partly due to the spatial association between road networks and settlement distribution in Bhutan’s valley-based settlement pattern. As settlements are typically located along accessible corridors, the settlement proximity layer captures a broader accessibility gradient that overlaps with road influence. Consequently, variations in road weights produce relatively smaller changes in suitability distribution compared to settlement proximity, which functions as the dominant accessibility driver.

The comparative sensitivity behavior of all factors is further illustrated in Section 6.3.

6.3 Factor Sensitivity Ranking

The ranked sensitivity results confirm a structured hierarchy of influence across all factors.

Rank	Factor	Variance	Average difference	Sensitivity based on variance	Sensitivity based on avg diff
1	Proximity to Settlements	10317.4	63.25	0.233	0.206
2	Tree Canopy Cover (TCC)	8113.01	57.29	0.183	0.186
3	Slope	7746.95	53.34	0.175	0.173
4	Proximity to Water	6913.12	45.83	0.156	0.149
5	Elevation (DEM)	6820.25	51.9	0.154	0.169
6	Proximity to Roads	4344.49	36.09	0.098	0.117
Total		44255.23	307.68	1	1

Table 6.3.1: Factor Sensitivity Ranking Based on Variance Analysis

The total variance of 44,255.23 and the normalized sensitivity sum of 1.00 confirm the internal consistency of the model and indicate stable system behavior under ±10% perturbation scenarios. The results demonstrate that the model remains robust despite moderate variations in factor weights.

Likewise, the graph below illustrates that proximity to settlements is the most influential factor within the model, whereas proximity to roads exhibits the lowest sensitivity under the tested weight variation scenarios. Detailed sensitivity analysis results are provided in Appendix–III.

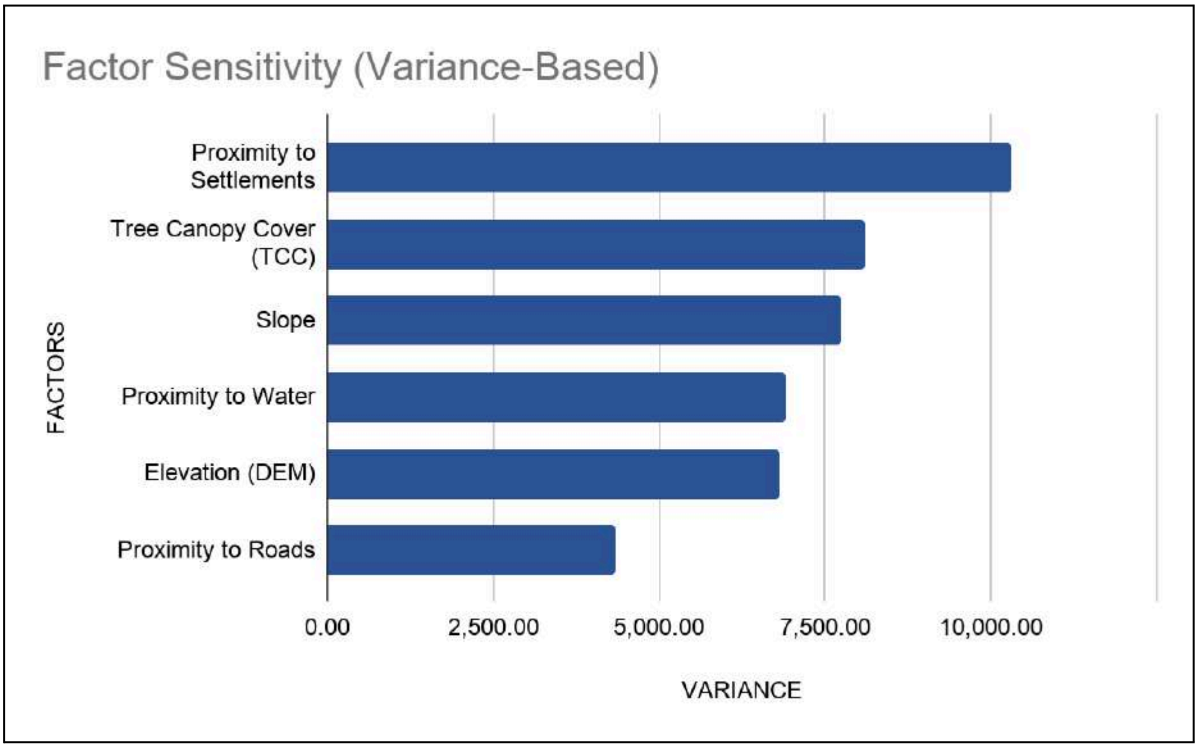

Figure 6.3.1: Factor Sensitivity Based on Variance from ±10% MCDA Weight Perturbation

6.4 Summary of Validation Findings

The sensitivity analysis demonstrates that the MCDA model is robust under moderate weight perturbations. Spatial overlay comparisons between baseline and sensitivity scenarios show that although localized variations occur, the overall spatial pattern of land suitability remains consistent.

The model is primarily driven by accessibility to settlements, followed by environmental constraints such as tree canopy cover and slope. Hydrological and topographic factors such as proximity to water and elevation show moderate influence on spatial variation.

Infrastructure-related factors such as roads exhibit comparatively lower sensitivity. This is likely due to partial redundancy within accessibility-related variables, where settlement proximity already represents a dominant spatial proxy for infrastructure accessibility in the model.

Overall, the validation confirms that the model is reliable for national-scale land suitability assessment and decision support.

7. LIMITATIONS

The **Identification of Alienable Land** at this national scale is a macro-level decision-support tool. The resulting baseline is subject to the following technical and operational limitations:

- **Preliminary Assessment and Field Validation Requirement:** The alienable land maps generated through this exercise provide a strategic overview and are intended strictly for preliminary assessment. Therefore, for actual land allocation, harmonization with existing land use plans, along with comprehensive field validation, is essential. As the MCDA model is based on secondary datasets, physical verification is required to confirm site-specific conditions such as soil stability, accessibility conditions, micro-encroachments, and community usage patterns that are not captured within the dataset framework. Desktop-verified information for highly suitable areas, together with sample field validation results, is presented in **Technical Report 2.**
- **Data Resolution Thresholds:** While the 20 meters spatial resolution is the highest available for MCDA, localized terrain features or micro-infrastructure smaller than the 20 meters grid cell may not be fully represented in the mapping.
- **Lack of Hazard Data Integration:** Due to the absence of uniform national spatial datasets, key localized hazard layers such as flood-prone areas, landslide susceptibility, and factor of safety were not incorporated into this suitability baseline.
- **Managed Forests and Traditional Rights:** Protected Areas (Core Zone and Transition Zone), Ramsar sites, and Biological Corridors are considered as constraints in the Final Net Baseline (Option 3). However, Community Forests (CF), Forest Management Units (FMU), and buffer and multiple-use zones of Protected Areas are not treated as strict constraints in the suitability analysis. In cases where land allocation is proposed within these areas based on the alienable land inventory, consultation with the Department of Forests and Park Services is required. In addition, swampy and marshy wetland areas were not fully delineated due to limitations in SAR (Synthetic Aperture Radar) based remote sensing, which may result in some areas classified as suitable actually being wetland terrain rather than truly suitable land.
- **Temporal Relevance:** The analysis is based on Land Use and Land Cover (LULC) 2020 data, while infrastructure and cadastral datasets reflect conditions as of September 2025. Rapid developments in the national road network and urban footprints will necessitate model updates every 2–3 years to maintain the accuracy of the identified alienable land.

8. CONCLUSION

This study establishes a scientific national baseline for the **Mapping of Alienable Land in Bhutan**, transitioning from theoretical potential to a constrained spatial reality.

- **The Land Gap:** Topographic, legal, and environmental constraints resulted in a 58% reduction of highly suitable (S1) land from the theoretical maximum.
- **Indicative Inventory:** The mapping identifies approximately 628,419.40 acres (S1) as high-priority "Search Areas."
- **The S2 Proposition:** While the Moderately Suitable (S2) category covers 2.4 million acres, its actual utility remains conditional and requires site-specific assessment, as suitability may vary depending on geotechnical, environmental, and ground conditions.
- **Decision Support:** This is a macro-level pre-screening tool, not a final land allocation. It serves to focus subsequent Manual Desktop Verification and Targeted Field Validation **(Technical Report 2)** on the most viable parcels.

9. RECOMMENDATIONS

9.1 Policy and Planning

- **Pre-screening of land demand:** Adopt the Option 3 baseline as the primary desktop filter for evaluating all state land allocation and lease requests.
- **Strategic Integration:** Embed the alienable land inventory into the Integrated Land Use and Spatial Planning framework.

9.2 Technical Enrichment

- **Data Integration:** Prioritize the inclusion of Forest Management Units (FMUs), Community Forests, and hazard datasets (flood, landslide, seismic) to refine land safety profiles.
- **Remote sensing enhancement:** Incorporate high-resolution satellite imagery and SAR (Synthetic Aperture Radar) data to improve identification of surface conditions such as water bodies and marshy or swampy terrain that may not be clearly represented in existing datasets.
- **Resolution Enhancement:** Continue improving spatial resolution as higher-quality Digital Elevation Models (DEM) become available.

9.3 Cross-sectoral Coordination and NSDI Utilization

- Promote the Bhutan **National Spatial Data Infrastructure (NSDI)** System as the central platform for standardized spatial data sharing and synchronization among relevant agencies to ensure consistent updates of infrastructure, environmental, and geospatial datasets. The **Centre for Geo-Information (CGI)** shall support this process as a key technical coordination entity where applicable, facilitating data harmonization and interoperability.

9.4 Validation and Future Directions

- **Verification Protocol:** Initiate systematic manual desktop verification and targeted field validation (Technical Report 2) to transition from an indicative mapping product to an operational Alienable Land Inventory.
- **Dynamic Update Cycle:** Establish a 2–3 year refresh cycle to ensure the dataset reflects rapid changes in national infrastructure and land use.
- **GeoAI Transition:** Transition the static baseline into an interactive Web-GIS dashboard integrated with GeoAI for real-time decision support in **Land Bank management System (LBMS)**

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APPENDIX-I: Technical Methodology for the Alienable Land Inventory

The following flowchart outlines the core technical workflow for the Alienable Land Inventory. Detailed explanations of each stage, including geoprocessing parameters and analytical logic, are provided in the sections that follow.

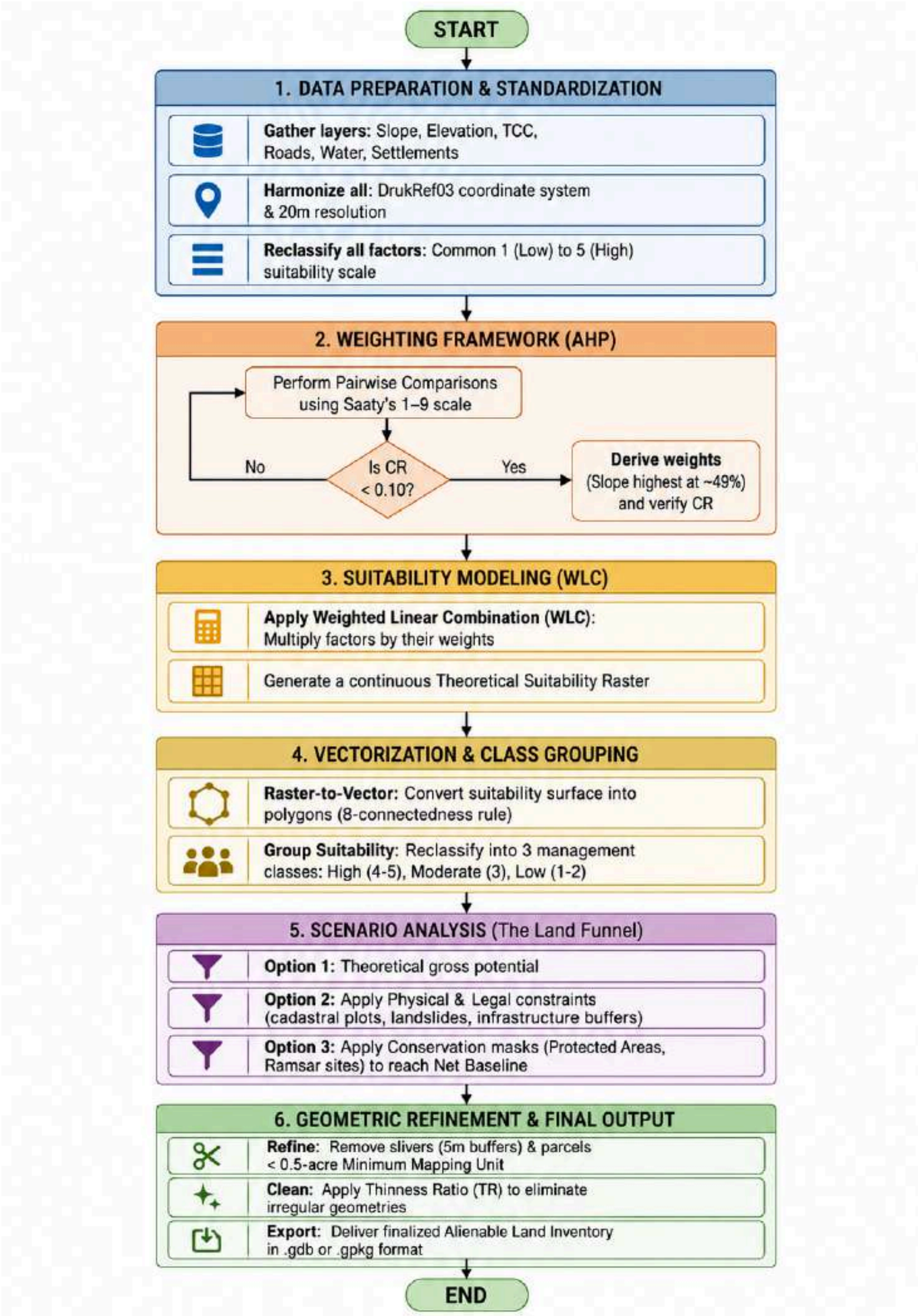


Figure Anx-I-1:Methodological Workflow for Alienable Land Identification

Section A: Data Inventory and Spatial Standardization

In Multi-Criteria Decision Analysis (MCDA), the structural integrity of the final output is inextricably linked to the quality and standardization of the input spatial layers. For a national-level assessment such as the Bhutan Alienable Land Inventory, rigorous spatial standardization is a strategic necessity to prevent "spatial drift"—pixel misalignment that can lead to significant geometric errors during overlay operations. By enforcing strict protocols on coordinate systems, resolution, and data types, we ensure that the mathematical operations performed during the weighting phase are executed on a unified, high-precision analytical grid.

A.1 Factor Identification and Data Provenance

The following table synthesizes the primary datasets used in this assessment, identifying their authoritative sources and their specific relevance to the land suitability framework.

Sl No.	Factor Name	Original Data Source (Agency)	Analytical Relevance
1	Slope	Cartosat DEM (NLCS)	Critical determinant for physical usability, construction cost, and mechanization potential.
2	Proximity to Settlement	Urban Boundary & NLUZ Rural Settlement (DHS/NLCS)	Assesses social integration, pull factors, and proximity to essential services and utilities.
3	Proximity to Water	Dept. of Water (DoW) / NLCS Topo Base Map	Evaluates feasibility for domestic consumption and gravity-fed irrigation systems.
4	Proximity to Road	Dept. of Surface Transport (DoST)	Determines market connectivity, ease of infrastructure extension, and emergency access.
5	Tree Crown Cover	Dept. of Forest and Park Services (DoFPS)	Identifies forest density to balance development with national conservation mandates.
6	Elevation	Cartosat DEM (NLCS)	Defines agro-ecological zoning, physiological stress levels, and climatic suitability.

A.2 Standardization Parameters

To maintain a coherent analytical environment, all datasets were processed using the following technical specifications:

- **Coordinate Reference System (CRS):** EPSG 5266 (DRUKREF 03 / Bhutan National Grid).
- **Spatial Resolution:** 20 meter pixel size, consistent with national planning requirements.
- **Data Type:** Float32. This is essential to prevent rounding errors during the AHP weighting phase, where small decimal weights (e.g., 0.0347) are multiplied against pixel values.
- **Rasterization Protocol:** Vector-to-raster conversions for water and settlement layers utilized a "Burn Value" of 1 to ensure binary presence is accurately captured before distance modeling.
- **Cell Alignment:** All layers were snapped to a common origin (125511.986, 3141841.656) and extent to ensure perfect vertical pixel alignment.
- **Raster Width & Raster Height:** 16807 & 9788
- **Spatial Extend:** National Bhutan Extend
-

A.3 Factor Rationale and Reclassification

To enable mathematical integration of heterogeneous variables, all factors were standardized to a common ordinal suitability scale ranging from 1 to 5.

Reclassification Value	Suitability Interpretation
5	Very High Suitability
4	High Suitability
3	Moderate Suitability
2	Low Suitability
1	Very Low / Unsuitable

1. Slope

Slope is the primary physical constraint in Bhutan's mountainous terrain. Gentle slopes (<10°) are highly suitable for development and agriculture, while slopes exceeding 30–35° are widely considered unsuitable due to erosion risk, slope instability, and high infrastructure costs. These thresholds follow established land evaluation guidelines and terrain suitability practices. .

Min Value (°)	Max Value (°)	Reclassification Scale	Suitability Remarks
0	10	5	Very High Suitability
10	15	4	High Suitability
15	25	3	Moderate Suitability
25	35	2	Low Suitability
> 35		1	Very Low/Unsuitable

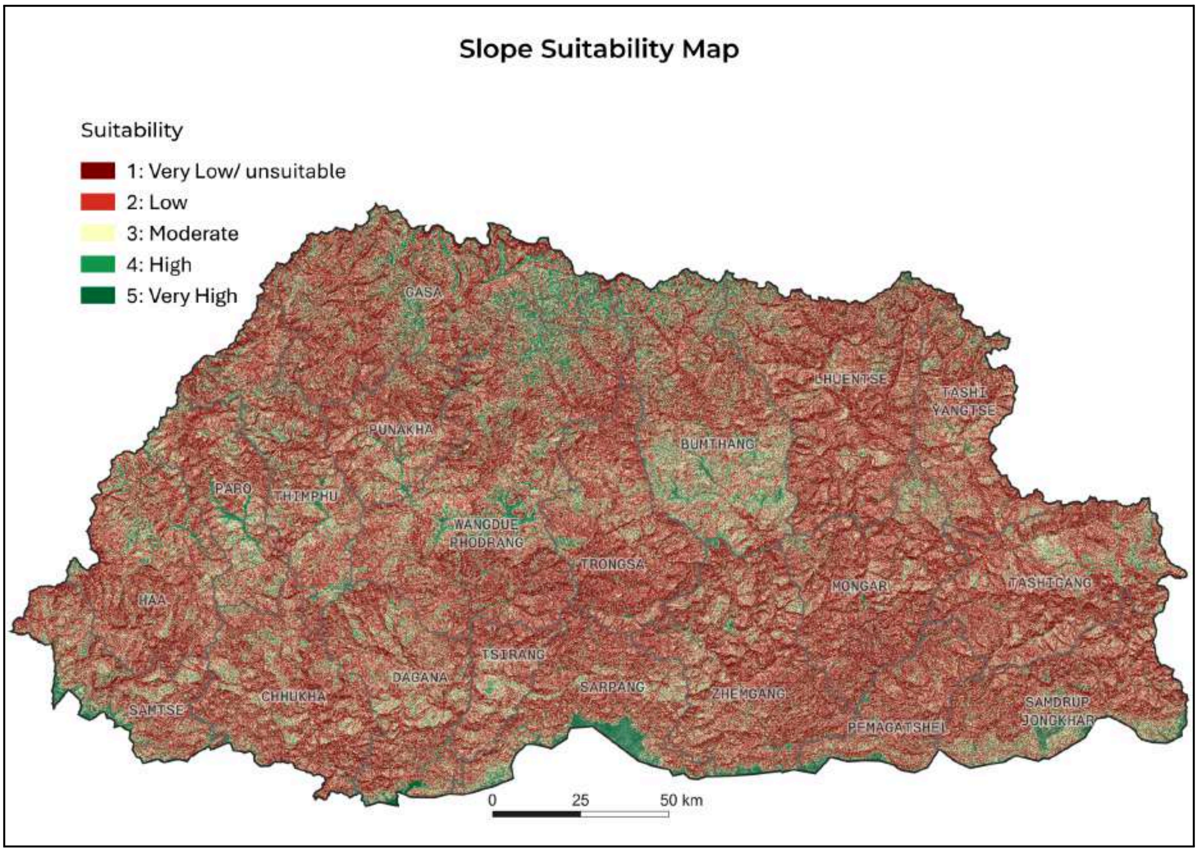


Figure Anx-I-2: Slope Suitability Map

2. Elevation

Elevation defines agro-ecological suitability, influencing climate, agricultural productivity, and human habitation. Lower elevations provide favorable conditions for settlement and agriculture, while higher elevations (>3000–3600 m) are constrained by low temperatures, reduced productivity, and environmental hazards typical of alpine regions.

Min Value(m)	Max Value(m)	Reclassification Scale	Suitability Remarks
100	1200	5	Very High Suitability
1200	2400	4	High Suitability
2400	3000	3	Moderate Suitability
3000	3600	2	Low Suitability
> 3600		1	Very Low/Unsuitable

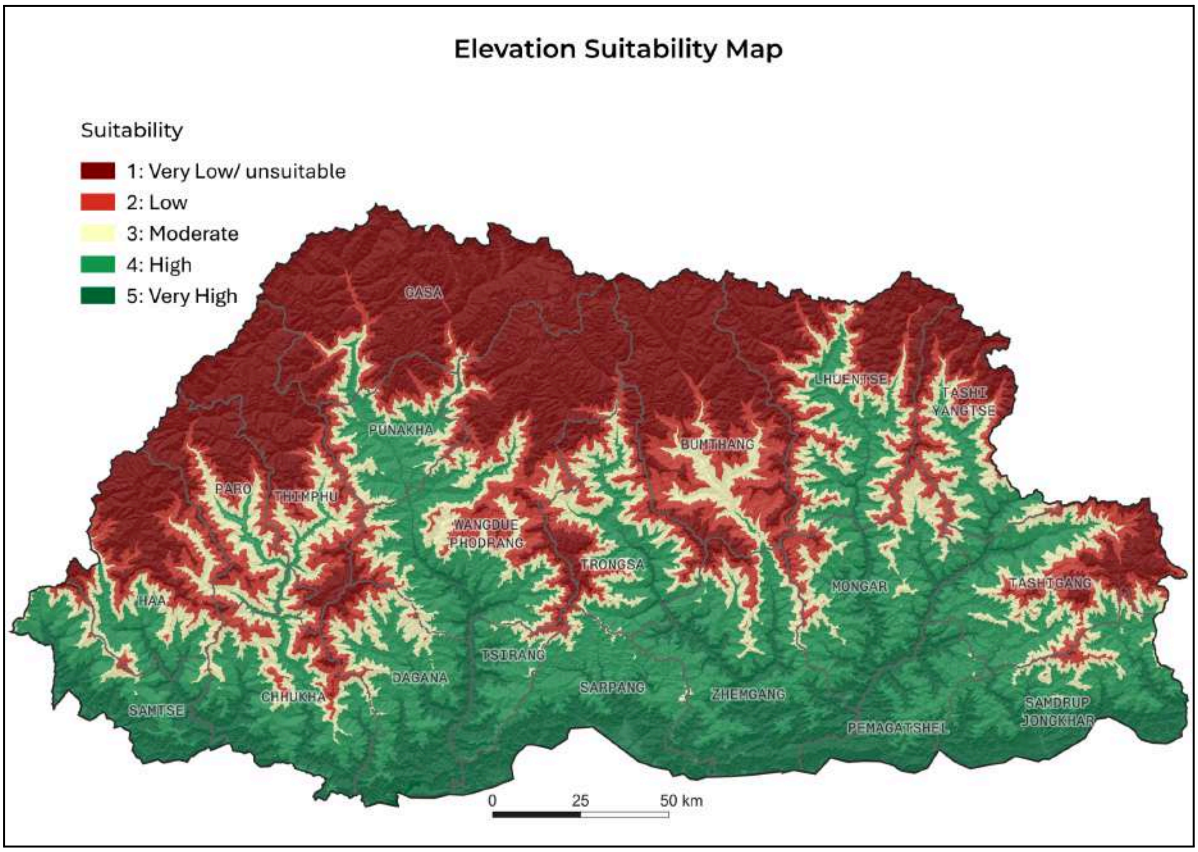


Figure Anx-I-3: Elevation Suitability Map

3. Proximity to Water

Water accessibility was modelled using a Euclidean distance surface with a distance decay principle, where suitability decreases as distance from water sources increases due to higher conveyance costs, reduced accessibility, and operational constraints. In Bhutan, optimal conditions are typically within 500–1000 m for gravity-fed systems, while distances beyond 2000–3000 m are generally limited by technical and economic feasibility.

Min Value(m)	Max Value (m)	Reclassification Scale	Suitability Remarks
0	500	5	Very High Suitability
500	1000	4	High Suitability
1000	2000	3	Moderate Suitability
2000	3000	2	Low Suitability
> 3000		1	Very Low/Unsuitable

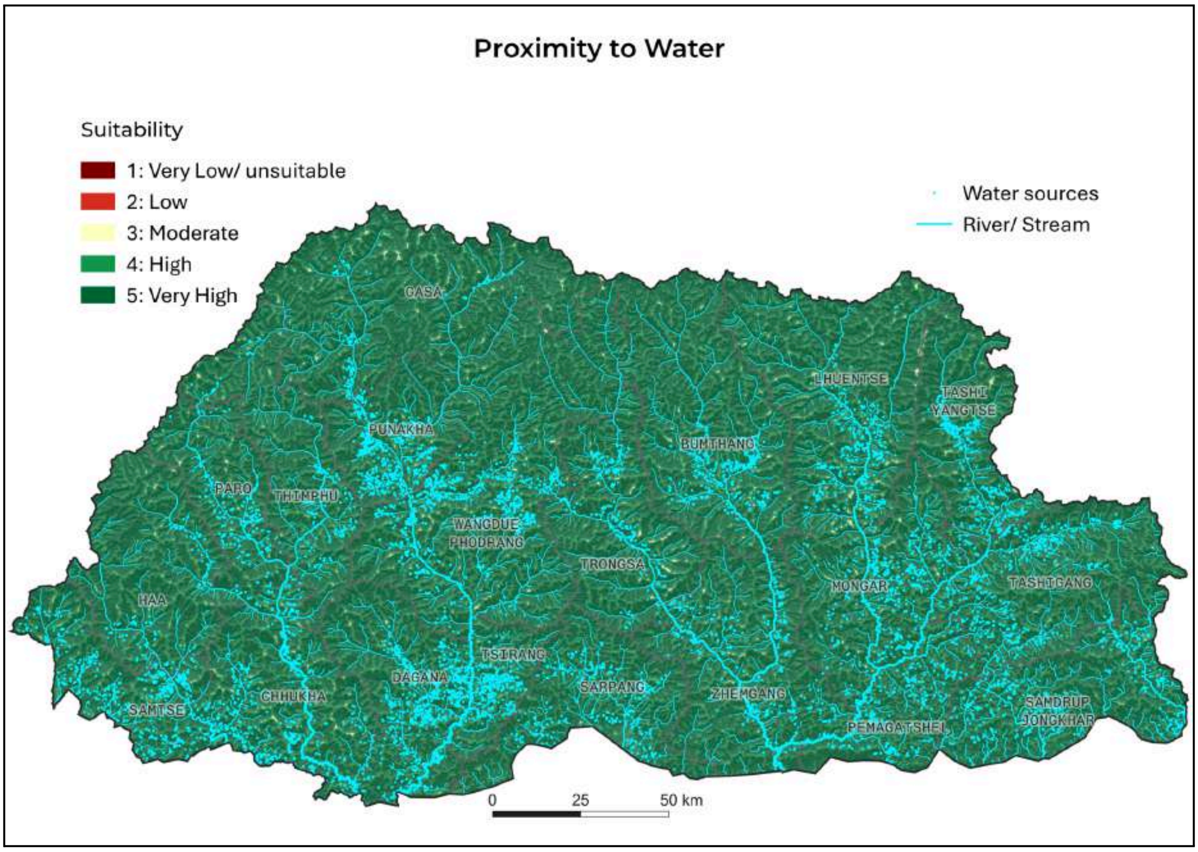


Figure Anx-I-4: Water Accessibility Map

4. Proximity to Road

Road proximity reflects infrastructure accessibility and economic connectivity. Areas near existing road networks are more suitable due to reduced transportation costs and ease of service delivery, whereas remote areas require significant investment for infrastructure expansion. Suitability therefore decreases progressively with distance from roads.

Min Value(m)	Max Value(m)	Reclassification Scale	Suitability Remarks
0	500	5	Very High Suitability
500	1000	4	High Suitability
1000	2000	3	Moderate Suitability
2000	3000	2	Low Suitability
> 3000		1	Very Low/Unsuitable

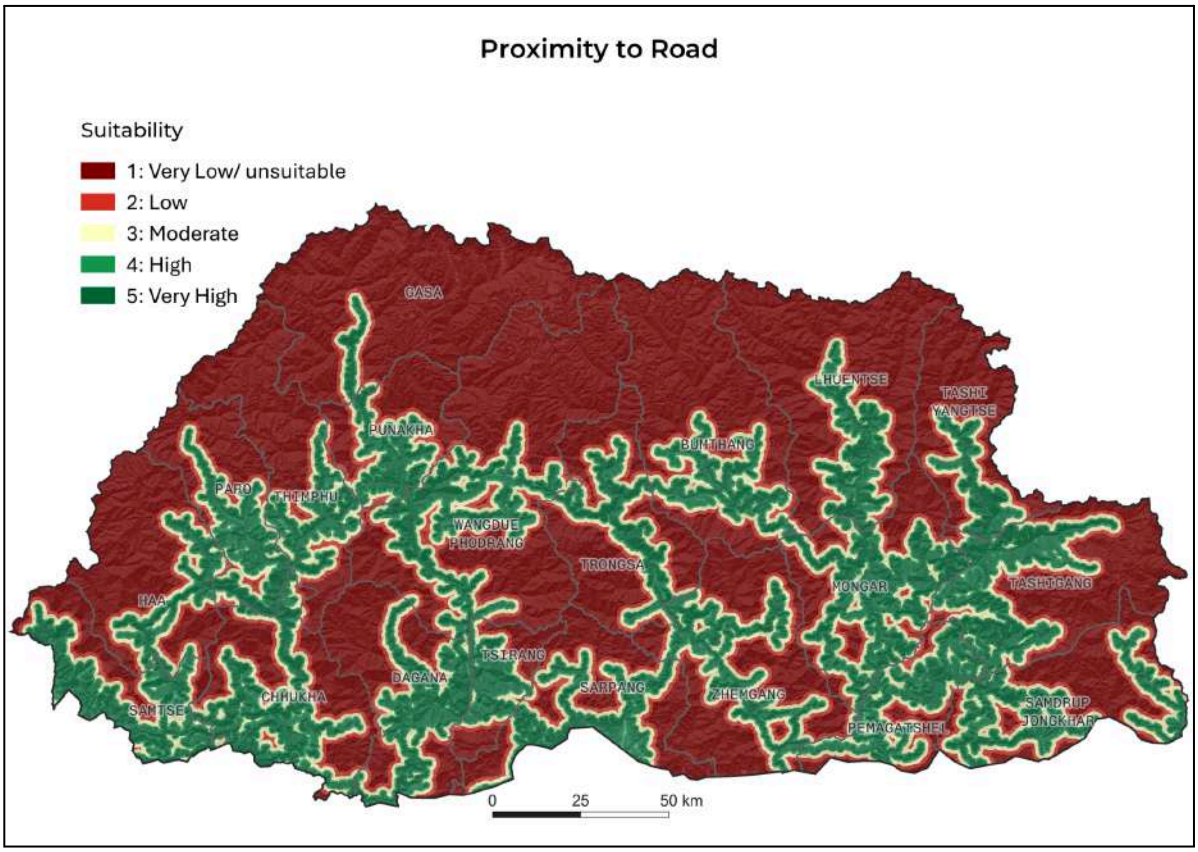


Figure Anx-I-5: Road Accessibility Map

5. Tree Crown Cover (TCC)

Tree Crown Cover was incorporated as an environmental constraint to guide development away from ecologically sensitive areas. Areas with low forest cover are considered more suitable, while dense forest areas are assigned lower suitability to minimize environmental impact and align with Bhutan's conservation priorities.

Min Value(%)	Max Value(%)	Reclassification Scale	Suitability Remarks
0	9	5	No Forest
9	39	4	Open Forest
39	59	3	Moderately Dense Forest
59	69	2	Dense Forest
69	100	1	Very Dense Forest

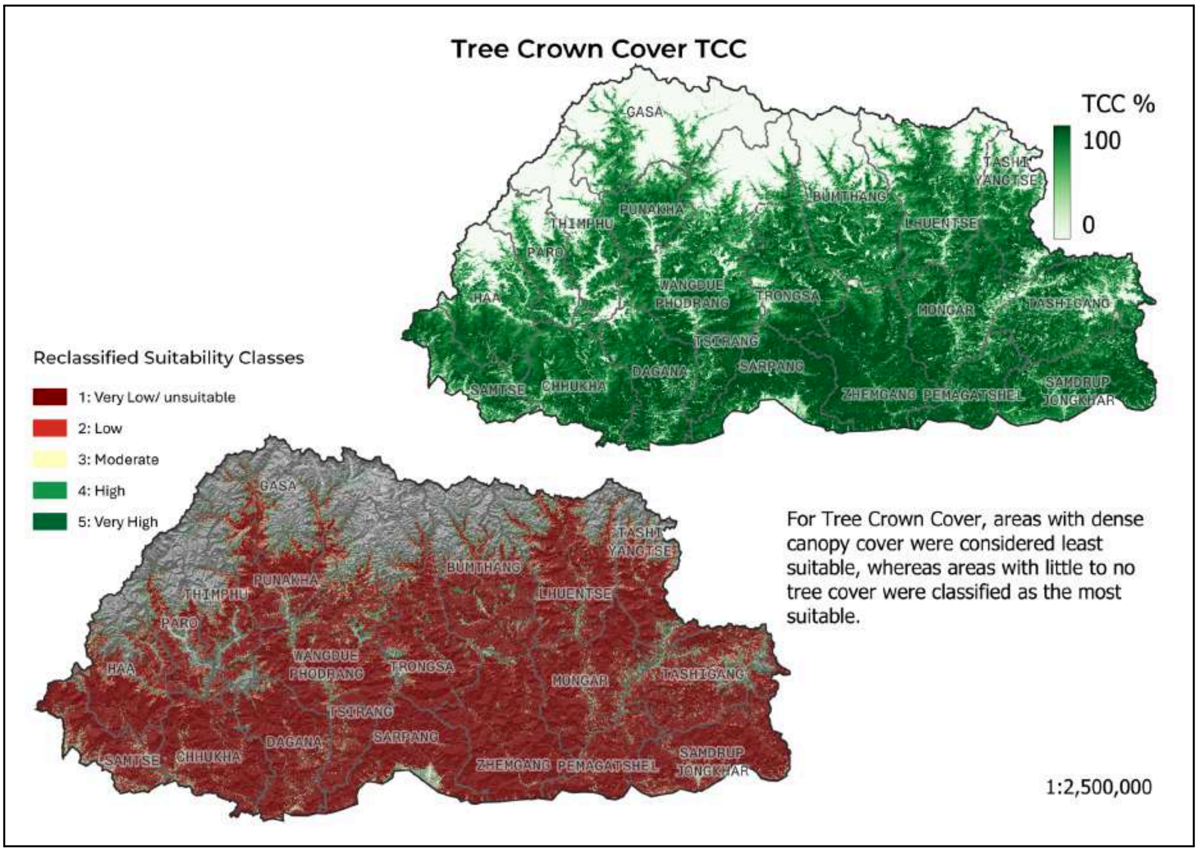


Figure Anx-I-6: Tree Crown Cover Suitability Map

6. Proximity to Settlement

Settlement proximity represents access to existing infrastructure, services, and social networks. Areas closer to settlements are more suitable due to integration with existing development, while distant areas are less favorable due to limited accessibility and higher service provision costs. .

Min Value (m)	Max Value(m)	Reclassification Scale	Suitability Remarks
0	500	5	Very High Suitability
500	1000	4	High Suitability
1000	2000	3	Moderate Suitability
2000	3000	2	Low Suitability
> 3000		1	Very Low/Unsuitable

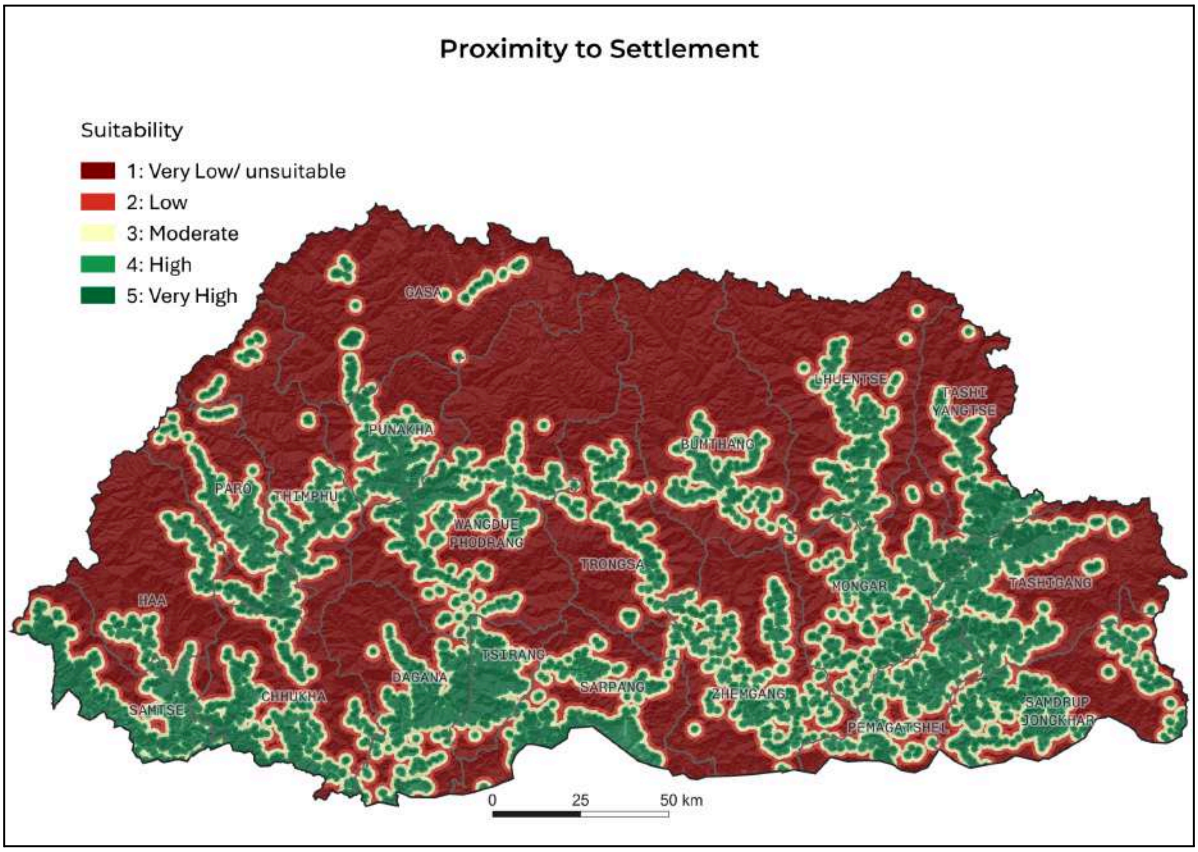


Figure Anx-I-7: Settlement Accessibility Map

These standardized layers form the basis for the weighting phase, where their influence is mathematically integrated using criteria adapted to Bhutan's conditions and established land evaluation frameworks.

Section B: Analytic Hierarchy Process (AHP)

The Analytic Hierarchy Process (AHP) is a structured multi-criteria decision-making (MCDM) technique used to transform qualitative expert judgment into quantitative weights through systematic pairwise comparisons. It is widely applied in spatial decision analysis due to its ability to ensure logical consistency using the Consistency Ratio (CR), which evaluates the coherence of judgments and improves the reliability of derived weights.

AHP decomposes complex land suitability problems into hierarchical comparisons, thereby enhancing transparency and reducing subjectivity in spatial weighting processes.

B.1 Methodological Rationale

AHP was selected due to its integrated internal validation mechanism. The Consistency Ratio ensures that expert judgments remain logically transitive and mathematically consistent. For example, if Slope is preferred over Road Proximity and Road Proximity over Elevation, AHP enforces a consistent hierarchical relationship between Slope and Elevation in the final weight structure, ensuring decision integrity.

B.2 AHP Preference Scale

Pairwise comparisons were conducted using Saaty's standardized 1–9 scale:

Intensity	Definition	Interpretation
1	Equal importance	Criteria contribute equally
3	Moderate importance	Slight preference based on judgment
5	Strong importance	Clear dominance of one criterion
7	Very strong importance	Significant dominance established
9	Extreme importance	Absolute dominance in decision context
2,4,6,8	Intermediate values	Compromise between adjacent judgments

B.3 Criteria Prioritization (Factor Ranking)

Criteria were prioritized based on their influence on alienable land suitability in Bhutan's mountainous terrain:

Slope > Settlement Proximity > Water Proximity > Road Proximity > Tree Crown Cover > Elevation

Slope is the dominant factor due to its direct influence on construction feasibility, slope stability, erosion risk, and engineering cost. Settlement and water proximity represent secondary accessibility drivers, while road proximity provides infrastructural connectivity. Tree Crown Cover and elevation act as tertiary environmental constraints that refine spatial suitability outputs.

B.4 AHP Weight Derivation

A reciprocal pairwise comparison matrix was constructed following Saaty's methodology. Each criterion was compared against all others to quantify relative importance. The matrix was normalized column-wise, and the mean of each row was used to derive the principal eigenvector.

Pairwise Comparison Matrix

Factor	Slope	Settlement	Water	Road	TCC	Elevation
Slope	1	4	5	6	8	9
Settlement	1/4	1	2	3	6	7
Water	1/5	1/2	1	2	4	5
Road	1/6	1/3	1/2	1	3	4
TCC	1/8	1/6	1/6	1/3	1	2
Elevation	1/9	1/7	1/5	1/4	1/2	1

B.5 Final Criterion Weights

Factor	Weight	Contribution (%)
Slope	0.485	48.5
Settlement Proximity	0.213	21.3
Water Proximity	0.136	13.6
Road Proximity	0.093	9.3
Tree Crown Cover	0.043	4.3
Elevation	0.031	3.1
Total	1	100

The results indicate that slope is the dominant controlling factor, contributing nearly half of the decision weight. Accessibility-related factors (settlement, water, and road proximity) collectively represent the secondary decision layer, while environmental variables (TCC and elevation) serve as refinement constraints in spatial suitability modeling.

B.6 Consistency Verification

To ensure reliability of expert judgment, AHP consistency was evaluated using the eigenvalue-based approach.

Weighted sum vectors were computed as:

λ_i = (Weighted Sum Vector$_i$) / w_i

Eigenvalue Results

Factor	λ Value
Slope	6.5822
Settlement	6.3588
Water	6.2292
Road	6.0717
TCC	5.9479
Elevation	6.0997

λ_{max} = 6.2149

B.7 Consistency Metrics

CI = (λ_{max} − n) / (n − 1) = 0.0430
RI (n = 6) = 1.24
CR = CI / RI = 0.0430 / 1.24 = 0.0347

B.8 Model Reliability

The Consistency Ratio (CR = 0.0347) is significantly below the acceptable threshold of 0.10, confirming strong internal consistency in expert judgments. This validates the robustness of the derived weights for integration into the GIS-based Multi-Criteria Evaluation (MCE) model.

B.9 Synthesis

The AHP results reflect the dominant influence of terrain conditions in Bhutan’s land suitability context, where slope governs physical feasibility, while accessibility and environmental variables provide secondary spatial refinement.

Section C: Multi-Criteria Evaluation (MCE)

C.1 Weighted Linear Combination (WLC)

The Multi-Criteria Evaluation (MCE) was conducted using the **Weighted Linear Combination (WLC)** method within the *TerrSet* liberaGIS environment. WLC integrates multiple standardized criteria into a single continuous suitability surface using AHP-derived weights.

$$S = \sum_{i=1}^{n}(W_i \times X_i)$$

Where:

- S = final suitability score
- w_i = weight of the i^{th} criterion
- x_i = standardized value (score) of the i^{th} criterion
- n = number of criteria

This formulation ensures proportional contribution of each criterion according to its relative importance.

C.2 Model Implementation

All thematic layers were reclassified to a uniform suitability scale and integrated as raster inputs in the MCE module. Each layer was multiplied by its corresponding AHP weight and aggregated to generate a continuous suitability surface.

Model parameters:

Parameter	Description
Platform	TerrSet liberaGIS v.20.0.1 (MCE module)
Method	Weighted Linear Combination (WLC)
Input Layers	6 reclassified criteria
Weight Source	AHP eigenvector
Output	Continuous suitability raster
Resolution	20 m

C.3 Model Workflow and Output Generation

The MCE workflow consists of two main stages: (i) integration of standardized thematic layers and (ii) generation of the final suitability surface through weighted aggregation.

Stage 1

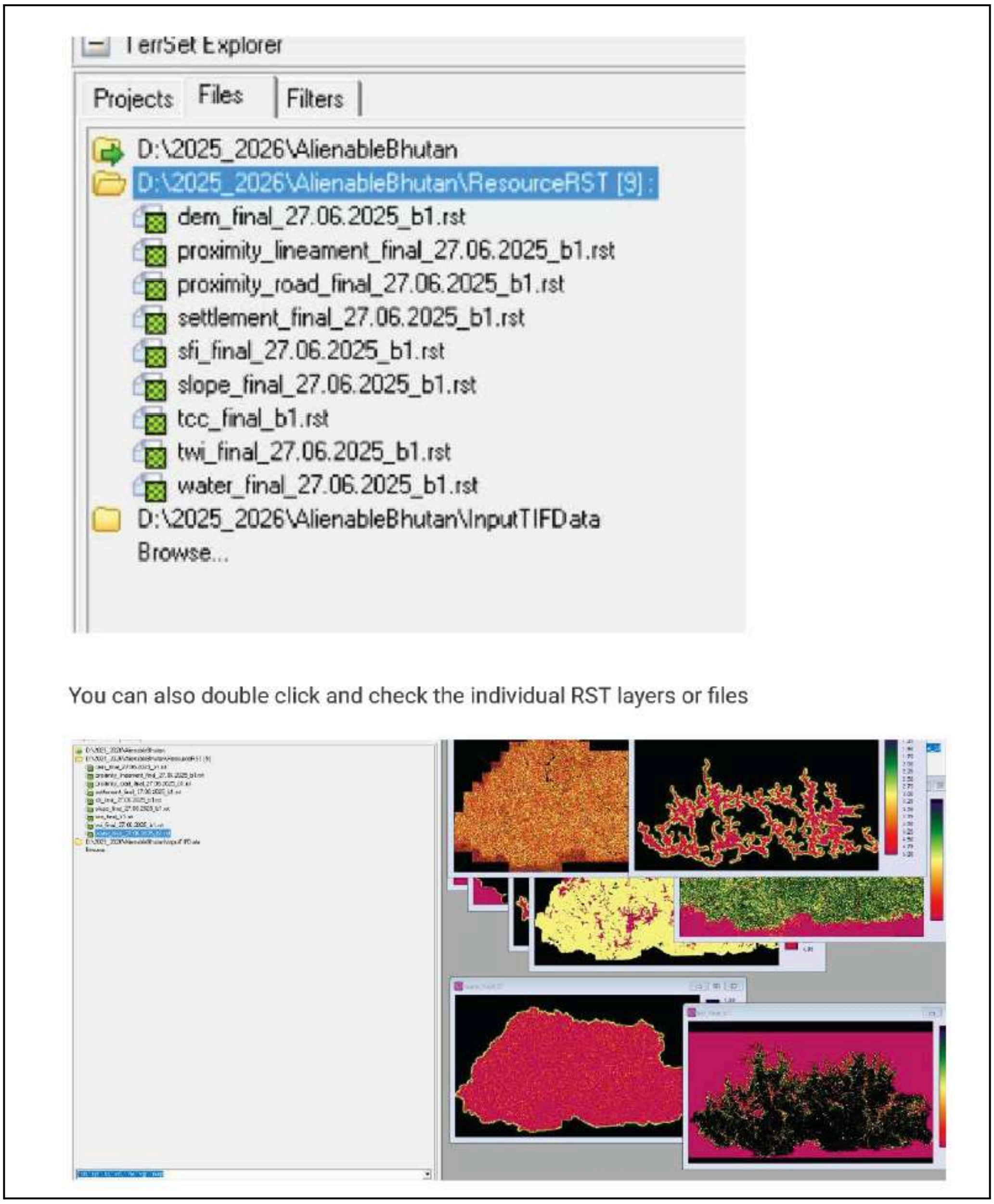


Figure Anx-I-8: Input Criteria Integration (MCE Preparation Stage)

This figure illustrates the integration of all standardized input layers used in the WLC model, including slope, settlement proximity, water proximity, road proximity, tree crown cover, and elevation.

Stage 2

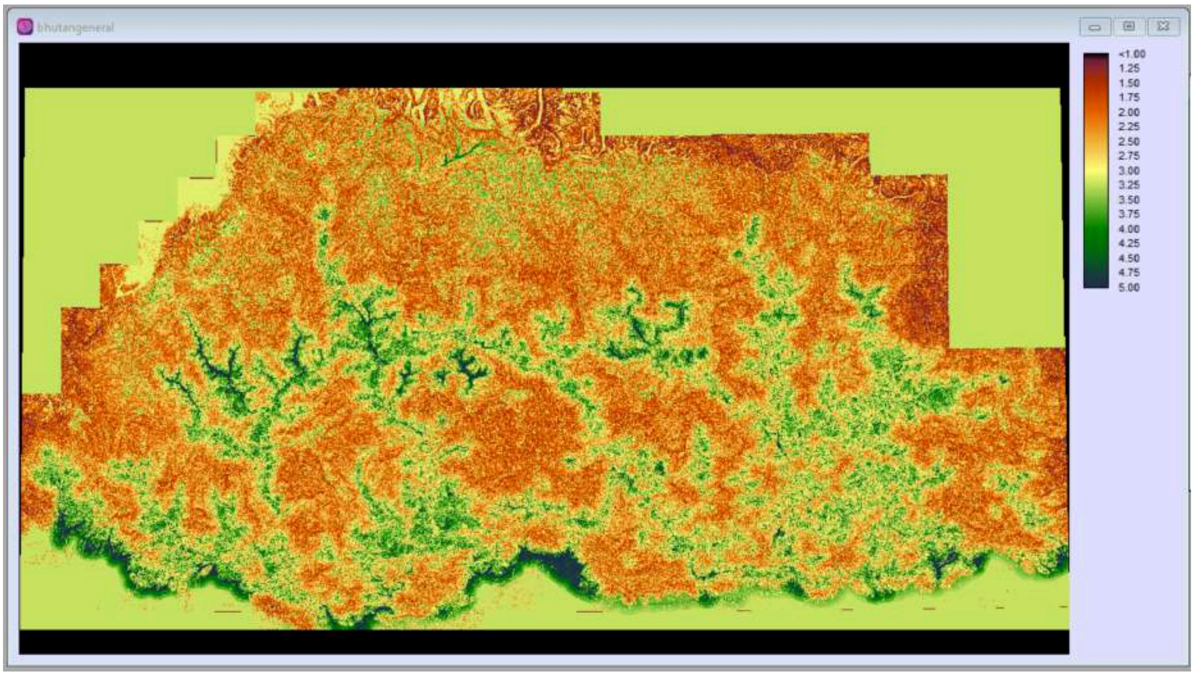


Figure Anx-I-9: Final Alienable Land Output Raster (.rst) in TerrSet liberaGIS

This figure presents the final spatial distribution of land suitability derived from the WLC model. The output represents a continuous suitability surface, where higher values indicate higher potential for alienable land development.

C.4 Output Interpretation

The final MCE output represents a spatially explicit suitability surface integrating terrain, accessibility, and environmental constraints. It provides a decision-support layer for identifying potential preliminary alienable land inventory and serves as the basis for subsequent classification into suitability classes.

C.5 Summary of MCE

The MCE framework integrates AHP-derived weights with GIS-based WLC modelling to produce a transparent and reproducible suitability surface. The workflow is validated through input-layer integration and final output visualization.

Section D: Post-Processing and Inventory Generation

The transition from theoretical suitability scores to actionable land parcels requires a shift from raster pixels to vector geometry, followed by the application of "land funnels"—successive layers of real-world constraints that filter out restricted or physically unsuitable areas.

D.1 Raster to Vector Conversion

To enable parcel-level interpretation and spatial decision-making, the 20 m resolution suitability raster was converted into vector polygons.

The process involved:

- Export of MCE outputs as high-precision GeoTIFF files
- Polygonization of raster cells into vector features in QGIS
- Application of an 8-connectedness rule, ensuring that diagonally adjacent cells with identical values were merged into continuous polygons, minimizing fragmentation and preserving spatial coherence

This conversion facilitated the transition from pixel-based suitability representation to decision-ready spatial units.

D.2 Suitability Reclassification Logic

The continuous suitability surface, with values ranging from 1 to 5, was reclassified into three operational categories to simplify interpretation and facilitate applied use in planning and institutional decision-making.

- **High Suitability:** DN values 4 and 5.
- **Moderate Suitability:** DN value 3.
- **Low Suitability/unsuitable:** DN values 1 and 2 (generally excluded from the final inventory).

This classification simplifies the continuous MCE output into discrete management zones suitable for policy and planning applications.

D.3 Land Funnel (Scenario Analysis)

A structured **land funnel approach** was applied to progressively refine gross biophysical suitability into a constrained and policy-compliant land inventory. Three scenarios were developed to represent increasing levels of restriction:

Scenario Name	Specific Constraints Applied	Planning Context
Option 1	No exclusion layers applied.	Gross potential land based purely on biophysical factors.
Option 2	Cadastral Boundaries, Road/River Buffers, and LULC Constraints (Landslides, Moraines, Rocky Outcrops, Sandy Banks, Snow, Glaciers, and Water Bodies).	Physical and legal constraints assessment.
Option 3	All Option 2 constraints plus Protected Areas and Ramsar-designated Wetlands.	Net baseline for alienable land, prioritizing conservation.

This stepwise filtering enables a transition from theoretical potential to policy-aligned developable land.

D.4 Data Refinement and Geometric Cleaning

A rigorous spatial refinement protocol was applied to ensure geometric integrity and eliminate processing artifacts.

Refinement Procedures

- **Sliver removal:** Small non-viable polygons were eliminated using buffer-based erase operations (5 m cadastral and 1 m footpath buffers).
- **Minimum mapping unit enforcement:** Polygons smaller than 0.5 acres were merged with adjacent dominant parcels using the *Eliminate (Largest Common Boundary)* method.
- **Shape validation:** The **Thinness Ratio** was computed to identify geometrically inefficient or elongated polygons:

$$TR = 4\pi \frac{A}{P^2}$$

Where:

- A = polygon area (m^2)
- P = perimeter (m)

Values closer to 1 indicate more compact and geometrically stable polygons, while lower values represent elongated or irregular shapes.

- **Topology validation:** Final datasets were checked for overlaps, gaps, and self-intersections to ensure a clean, error-free geodatabase structure.

D.5 Final Output and Inventory Generation

The finalized alienable land inventory was exported in standardized geospatial formats to support cross-agency interoperability, long-term data management, and integration within national GIS platforms.

Output Formats

- ESRI Geodatabase (*.gdb*)
- GeoPackage (*.gpkg*)

These formats preserve spatial attributes, topology, and metadata while ensuring compatibility across major GIS environments.

Final Alienable Land Inventory Visualization (Scenario Comparison)

The maps below illustrate the progressive reduction of developable land through successive application of physical, legal, and environmental constraints, demonstrating the transition from gross suitability to the final alienable land baseline.

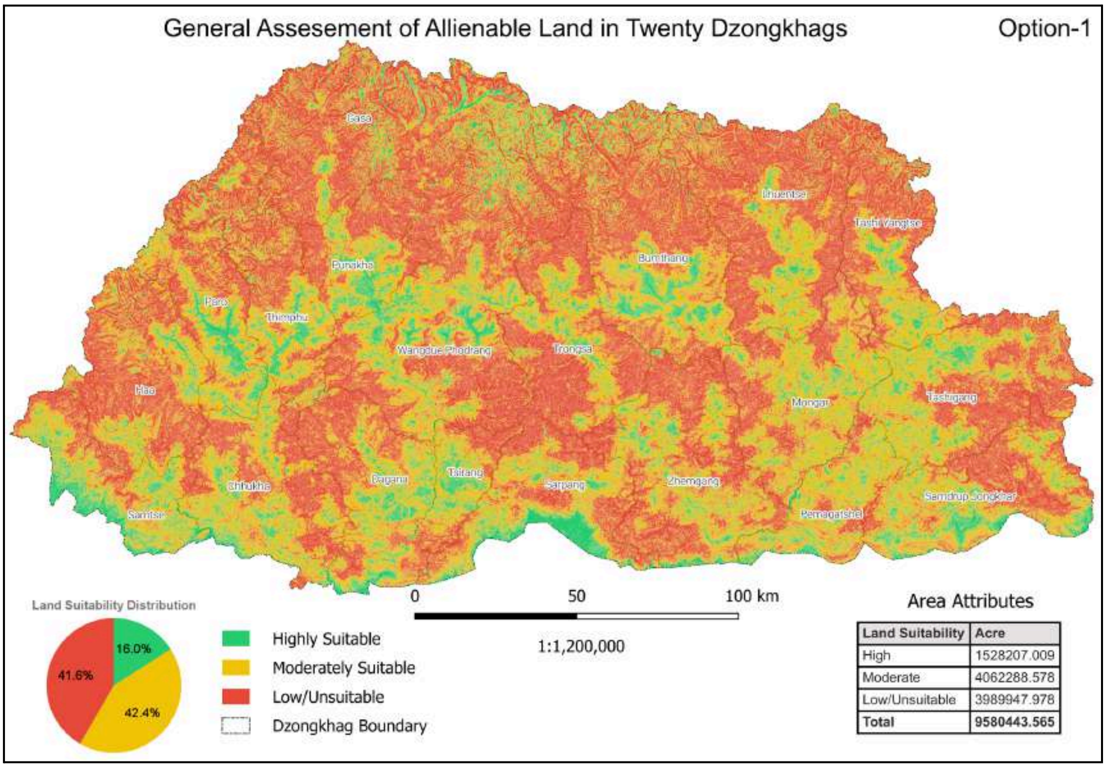


Figure Anx-I-10: Option 1: No exclusion layers applied.

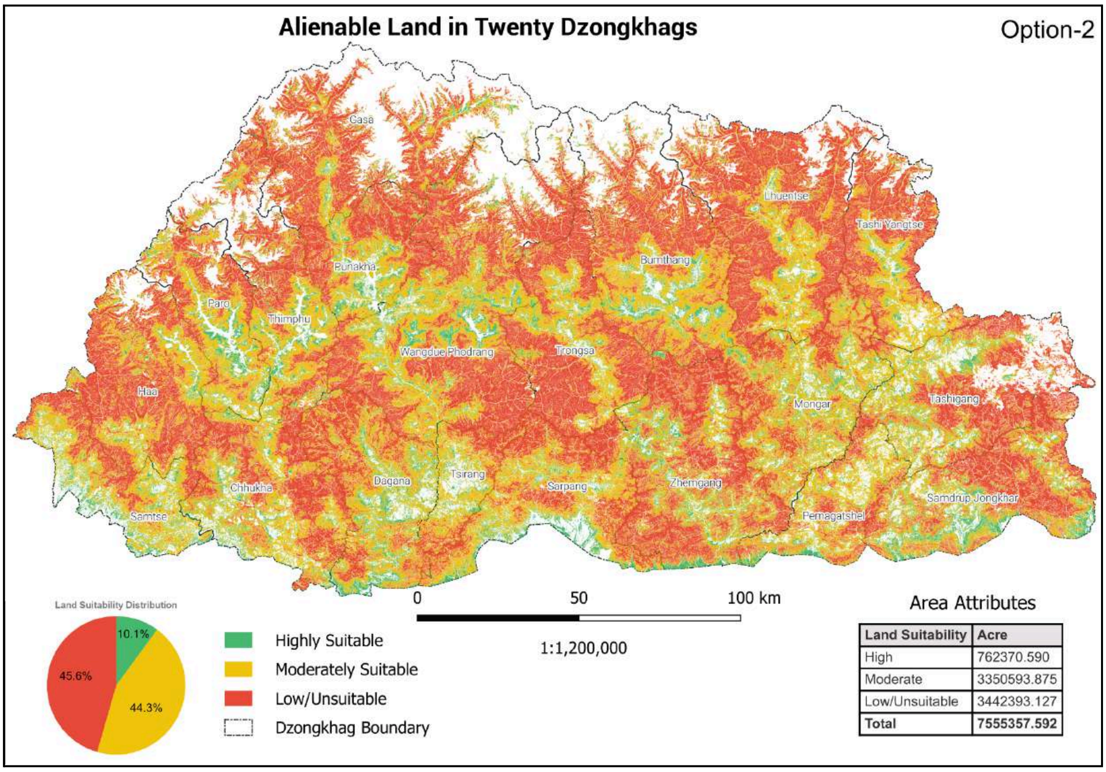


Figure Anx-I-11: Option 2: Physical and legal constraints assessment.

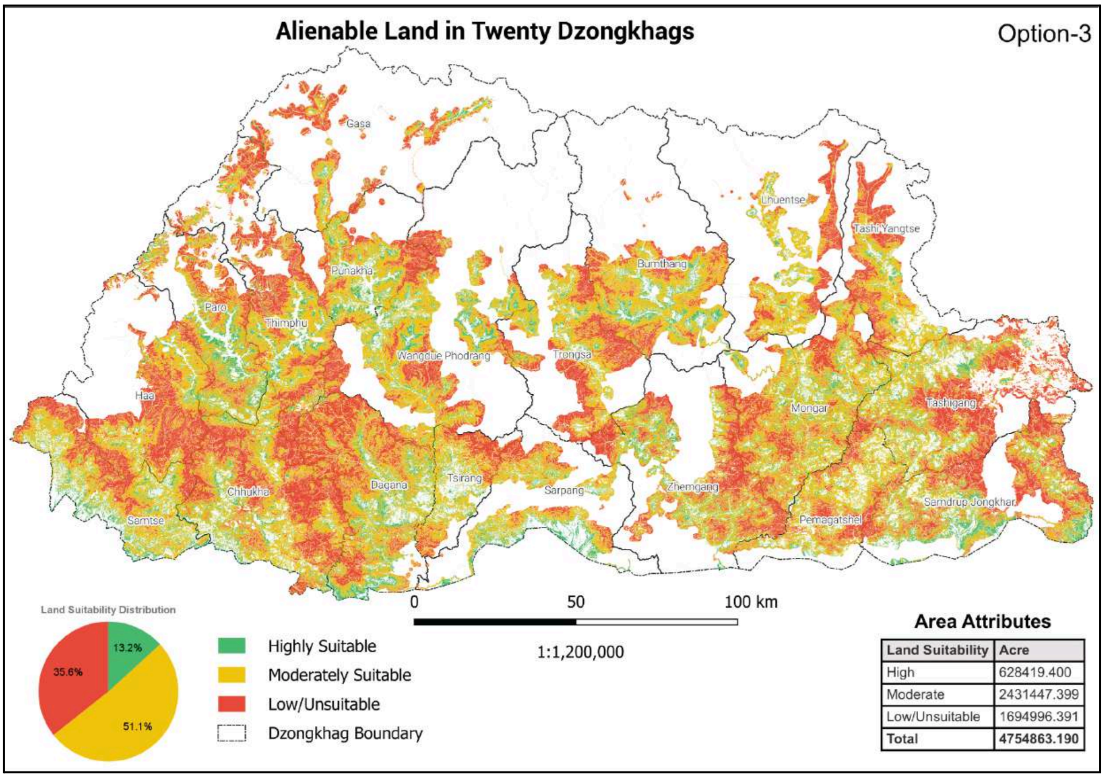


Figure Anx-I-12: Option 3: Net baseline for alienable land, prioritizing conservation.

Summary

The post-processing framework systematically transformed raster-based suitability outputs into vector based features. Integration of scenario-based constraint modelling, geometric refinement, and standardized geospatial outputs ensured that the final alienable land inventory is spatially robust, interoperable, and suitable for preliminary assessment for institutional planning and land management applications.

APPENDIX- II: Dzongkhag-wise Alienable Land Statistics and Scenario-wise Spatial Outputs

Section A: Dzongkhag-wise Suitability Contraction Across Scenario Options

This section presents Dzongkhag-wise land suitability distribution under three scenario options (Option 1–3), illustrating the progressive decline of alienable land under increasing spatial, legal, and environmental constraints.

Sno	Dzongkhag	Suitability Class	Option 1	Option 2	Option 3
1	Bumthang	High	104573.552	51109.434	40069.759
		Moderate	278775.418	215998.216	131241.062
		Low/Unsuitable	287397.462	221954.392	67562.758
2	Chhukha	High	81170.273	41396.568	41322.507
		Moderate	222282.298	198259.764	198226.312
		Low/Unsuitable	162278.511	156376.265	156376.623
3	Dagana	High	85245.569	46708.218	45282.573
		Moderate	190892.934	176700.984	167812.318
		Low/Unsuitable	149825.769	148735.292	138573.521
4	Gasa	High	93875.153	27507.219	12317.994
		Moderate	275307.402	129821.658	48807.364
		Low/Unsuitable	405581.816	223468.848	54971.610
5	Haa	High	42543.941	24150.382	17161.119
		Moderate	185744.491	162040.358	106027.261
		Low/Unsuitable	242238.313	225043.941	118170.968
6	Lhuentse	High	65307.247	28781.487	16788.804
		Moderate	264531.343	208097.770	84213.846
		Low/Unsuitable	377456.181	323035.071	51308.649
7	Monggar	High	78440.250	40515.658	37405.993
		Moderate	250733.645	227404.868	199087.579
		Low/Unsuitable	152090.888	151530.186	111155.249

8	Paro	High	67835.763	39473.215	37681.634
		Moderate	145899.428	125437.887	109355.522
		Low/Unsuitable	103913.222	84113.980	55937.588
9	Pemagatshel	High	58019.956	28445.164	21091.940
		Moderate	139974.750	116150.243	99451.515
		Low/Unsuitable	55060.833	52906.659	43403.874
10	Punakha	High	53077.034	29801.459	27801.594
		Moderate	110477.210	103534.181	81976.528
		Low/Unsuitable	110930.859	108576.579	44521.471
11	Samdrup Jongkhar	High	94515.475	50430.168	43076.894
		Moderate	208808.982	183329.913	144404.033
		Low/Unsuitable	159738.656	153058.824	99991.635
12	Samtse	High	103102.956	48542.604	48431.938
		Moderate	145777.403	129406.592	127855.671
		Low/Unsuitable	72840.239	71473.068	66201.701
13	Sarpang	High	94627.423	46282.665	34918.231
		Moderate	164196.705	152645.651	74461.918
		Low/Unsuitable	150254.340	148484.778	37429.252
14	Tashigang	High	82639.276	31919.320	31405.568
		Moderate	244817.422	175578.502	164348.755
		Low/Unsuitable	217314.084	172433.867	135121.492
15	Thimphu	High	64059.974	35814.256	33721.432
		Moderate	185928.378	141811.332	132366.985
		Low/Unsuitable	193977.014	135459.379	117101.081
16	TrashiYangtse	High	43383.296	18462.055	14709.391
		Moderate	150314.089	122236.836	80631.957
		Low/Unsuitable	164734.938	145283.914	61833.799
17	Trongsa	High	44814.036	24890.314	20755.240
		Moderate	165881.765	151084.200	99270.207
		Low/Unsuitable	236561.036	235333.456	92445.021
18	Tsirang	High	38250.284	16479.949	15415.507
		Moderate	73303.035	63631.852	55400.430

		Low/Unsuitable	45973.638	45686.173	35336.281
19	Wangdue Phodrang	High	155485.579	88510.312	58714.421
		Moderate	399244.908	330491.986	175375.122
		Low/Unsuitable	441943.620	381470.972	102419.261
20	Zhemgang	High	77239.973	43150.144	30346.860
		Moderate	259396.971	236931.084	151133.014
		Low/Unsuitable	259836.559	257967.483	105134.555

Section B: Dzongkhag-wise Suitability Maps (Atlas) and Comparative Analysis Across Options

The following pages present Dzongkhag-wise maps of alienable land suitability generated under different scenario options using the land funnel (constraint-based) approach. These maps illustrate how the spatial extent of alienable land changes as progressively stricter constraints are applied across Options 1 to 3.

In addition, statistical charts are provided to facilitate comparative analysis of land contraction across the different scenarios, clearly demonstrating the reduction in suitable land as additional constraints are introduced. The initial analysis was carried out in three phases, as illustrated in Figure Anx-II-21.

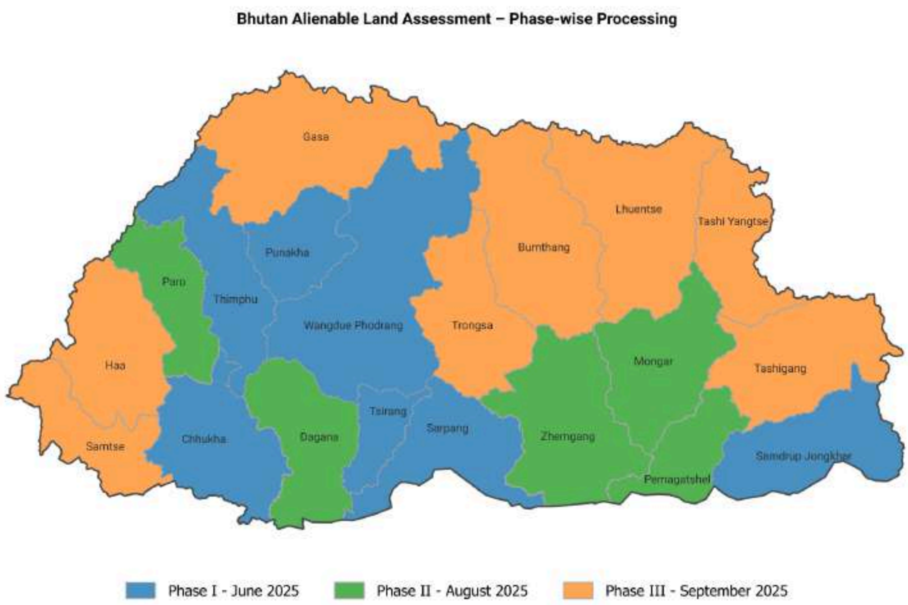


Figure Anx-II-1: Timeline of MCDA Analysis for Alienable Land Inventory

Bumthang

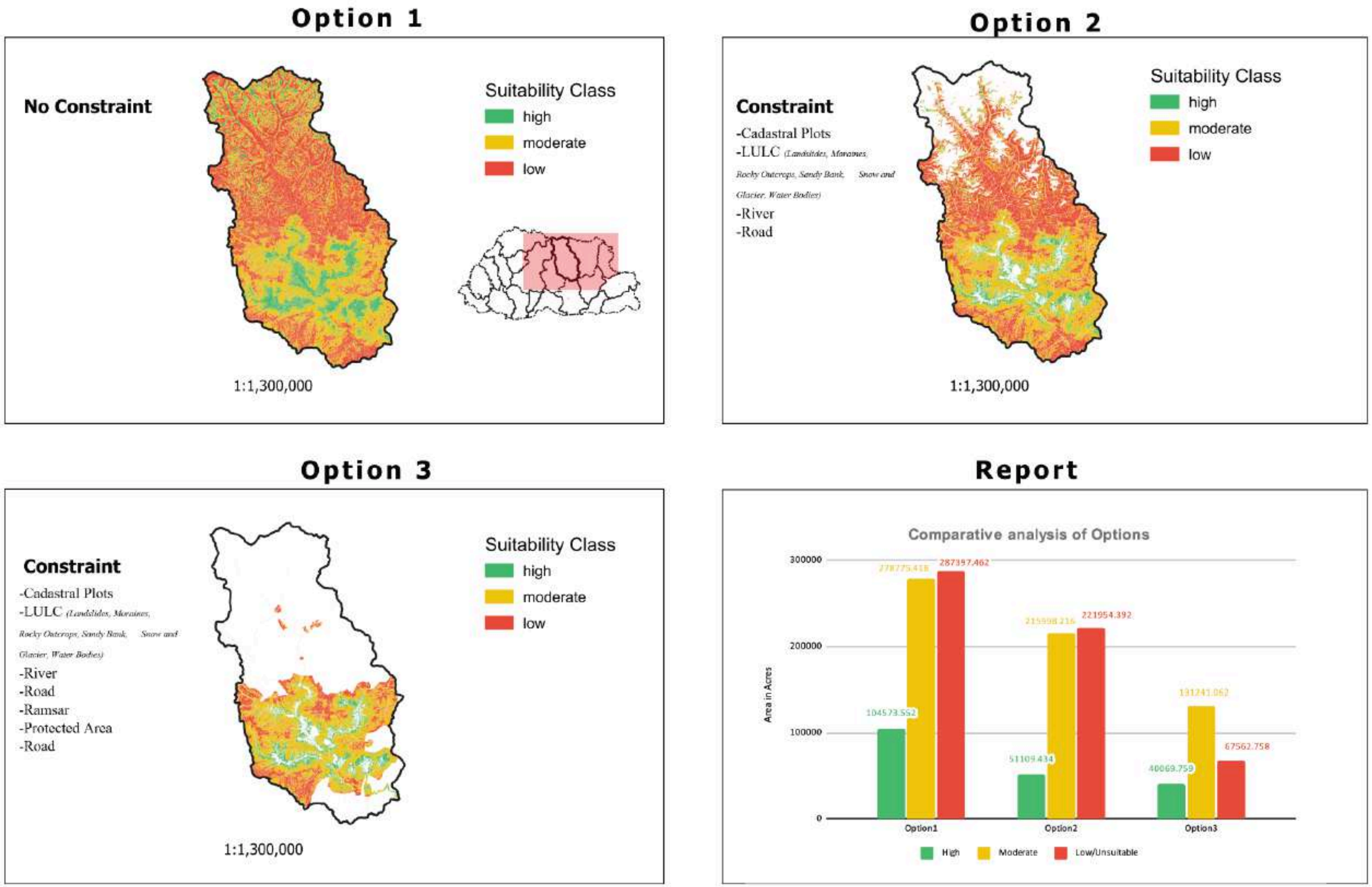


Figure Anx-II-2: Alienability Status of Bumthang Dzongkhag

Chhukha

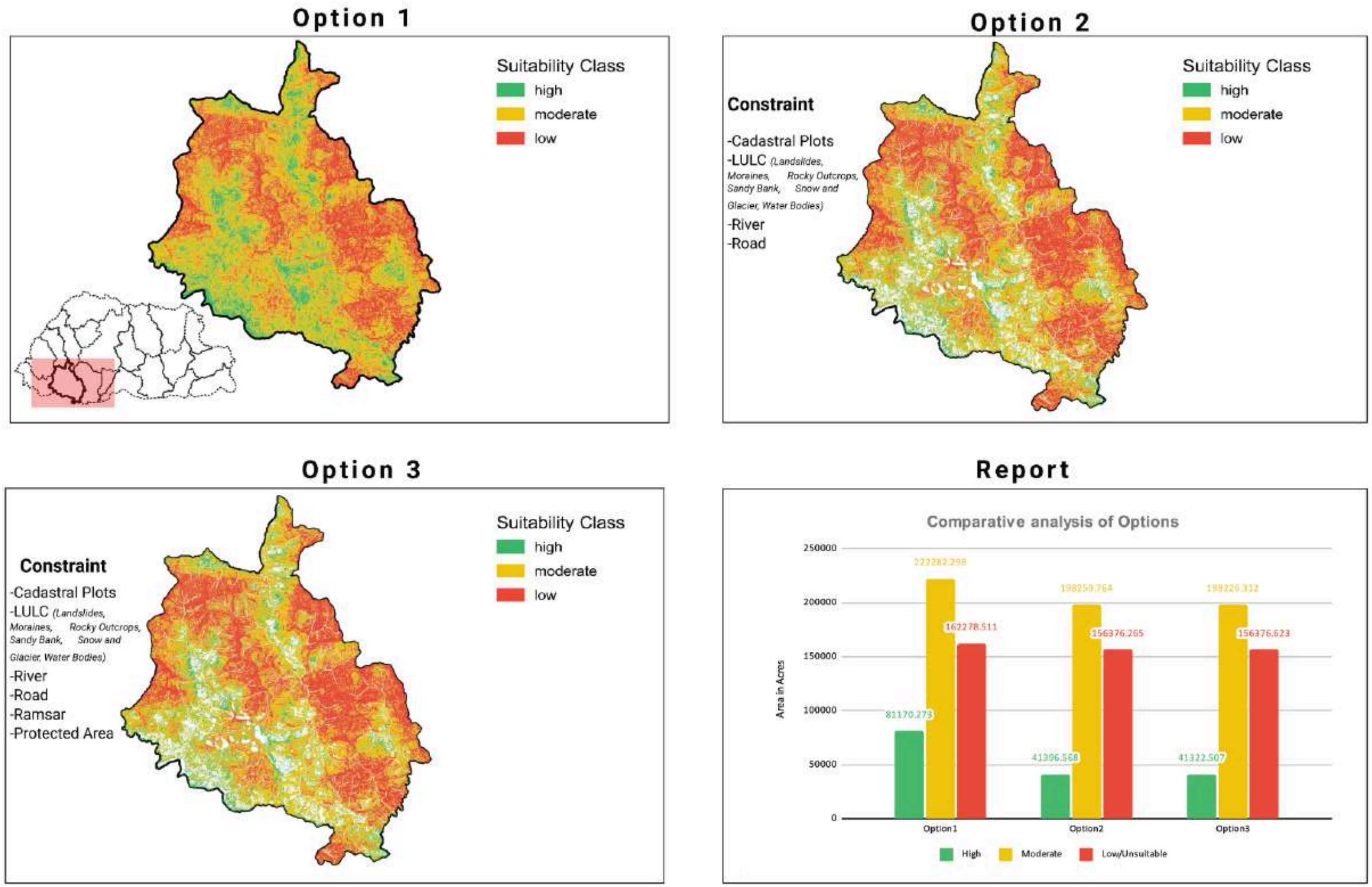


Figure Anx-II-3: Alienability Status of Chhukha Dzongkhag

Dagana

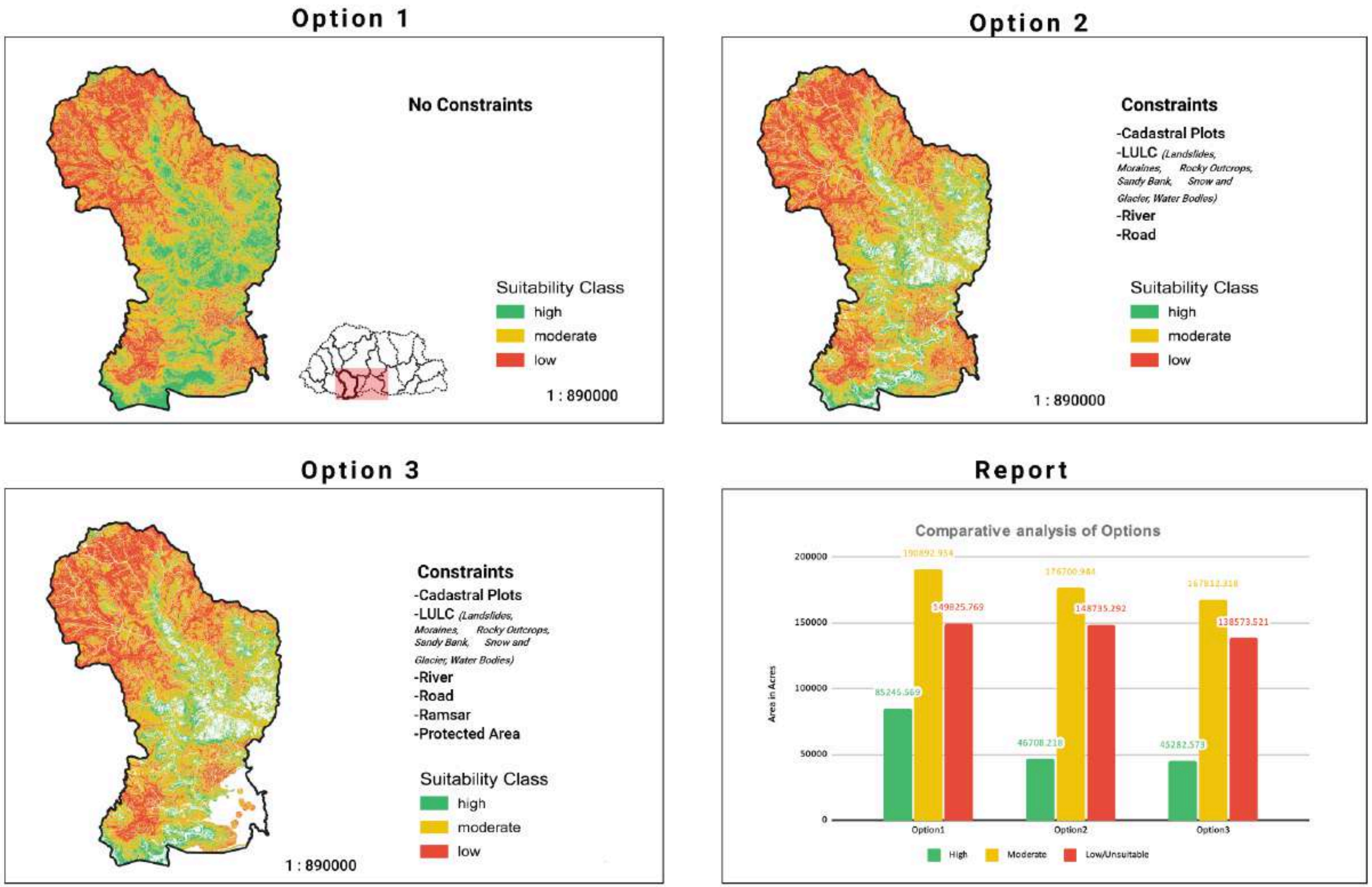


Figure Anx-II-4: Alienability Status of Dagana Dzongkhag

Gasa

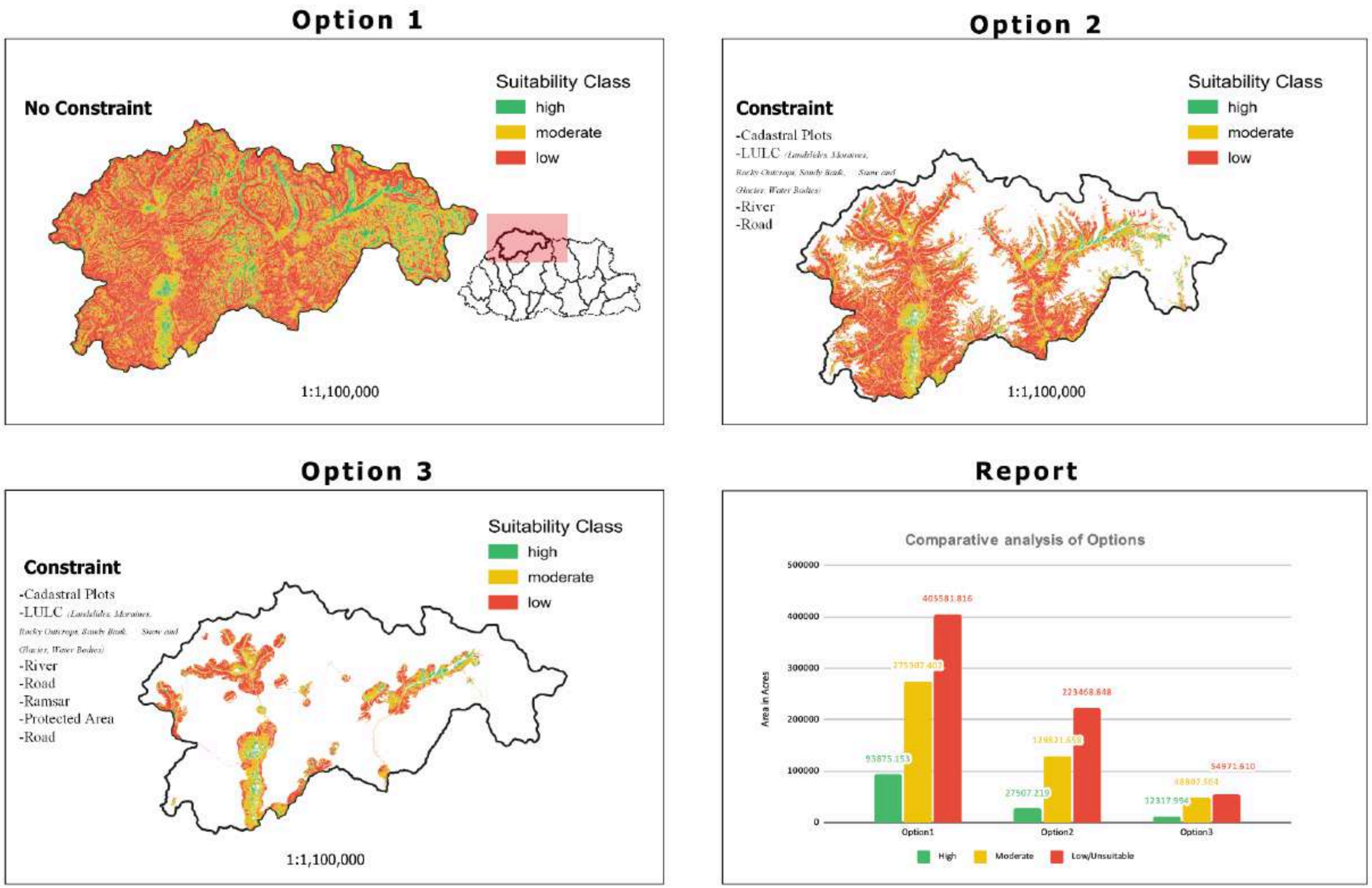


Figure Anx-II-5: Alienability Status of Gasa Dzongkhag

Haa

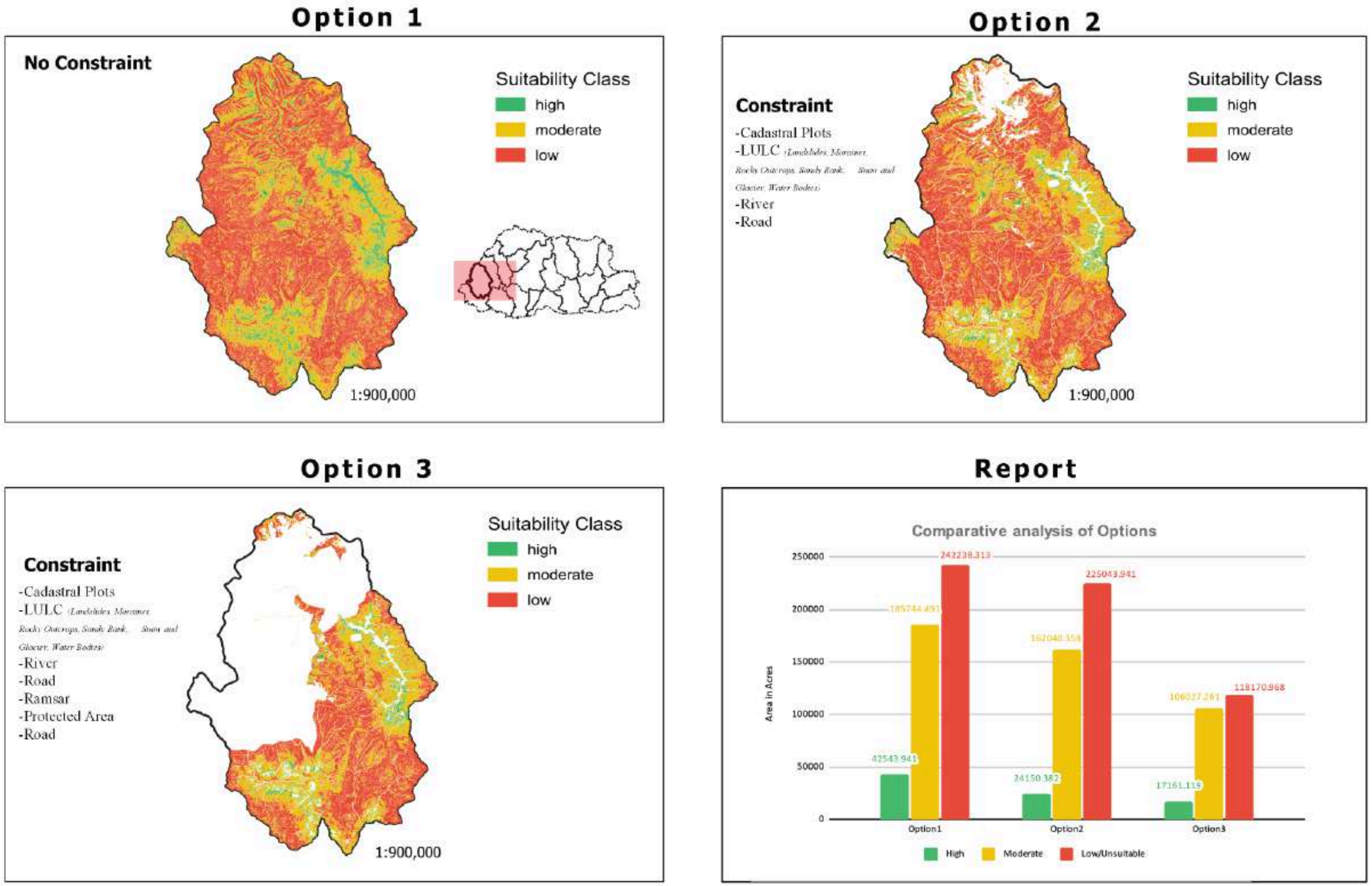


Figure Anx-II-6: Alienability Status of Haa Dzongkhag

Lhuentse

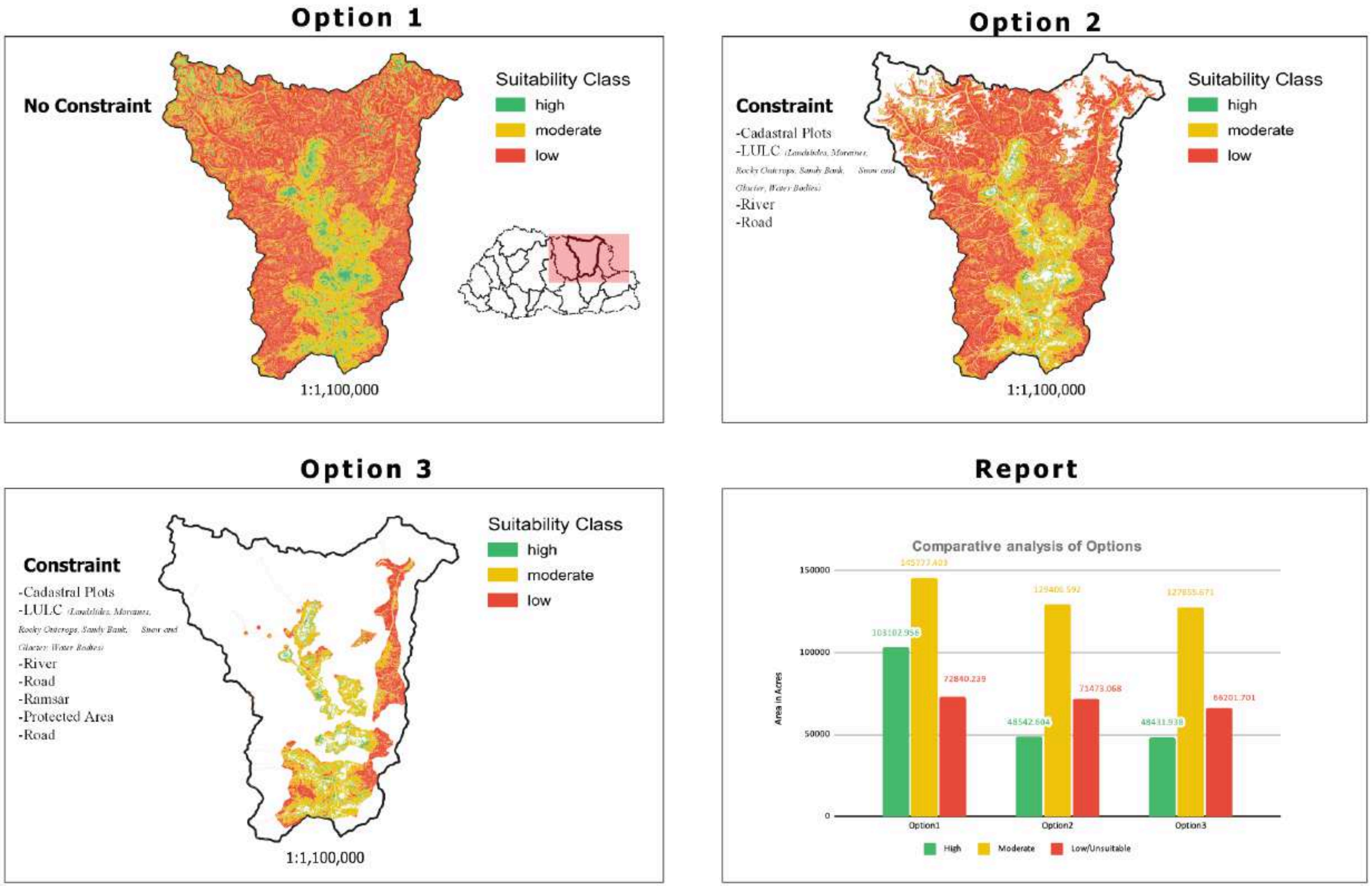


Figure Anx-II-7: Alienability Status of Lhuentse Dzongkhag

Mongar

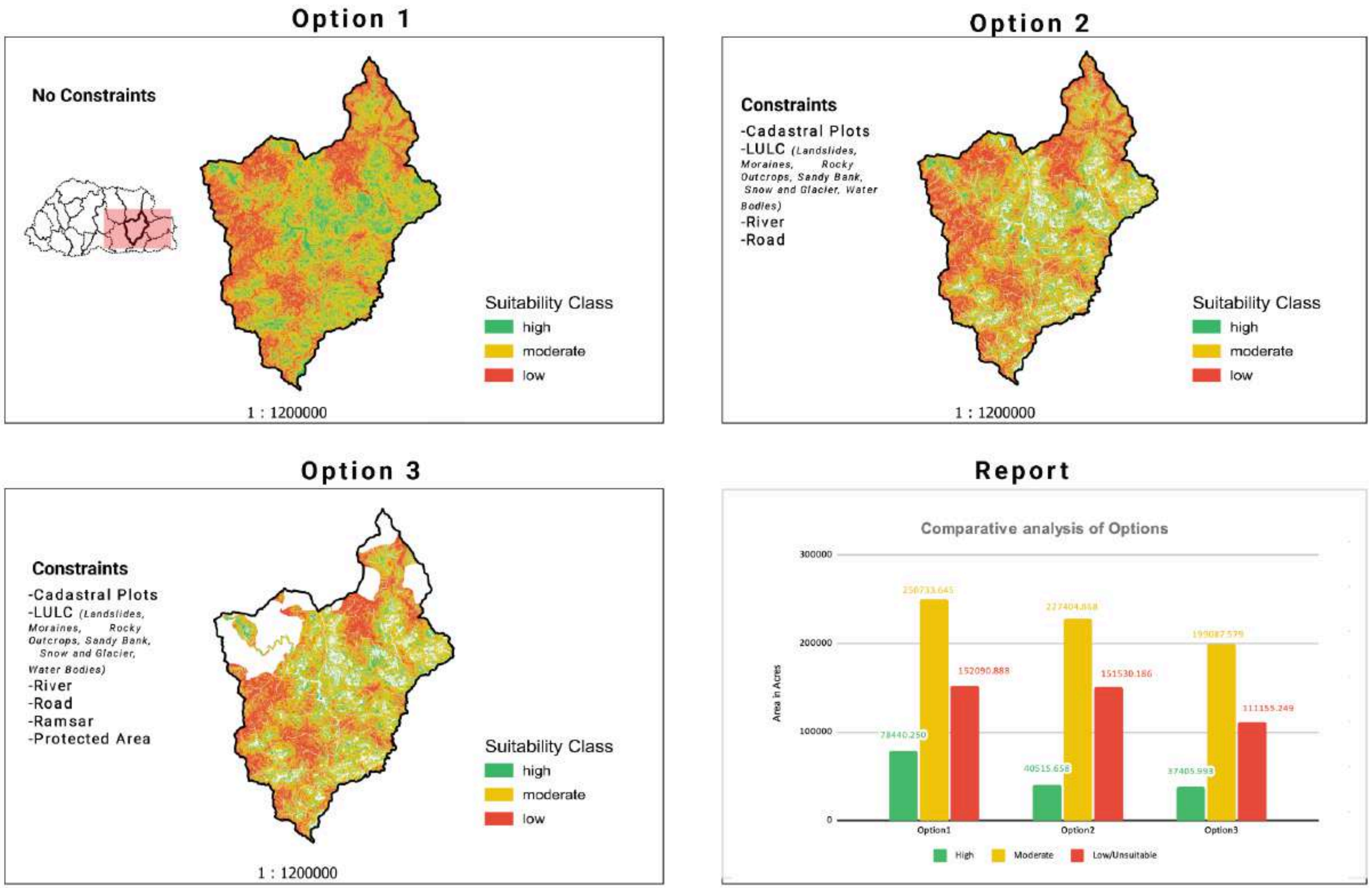


Figure Anx-II-8: Alienability Status of Mongar Dzongkhag

Paro

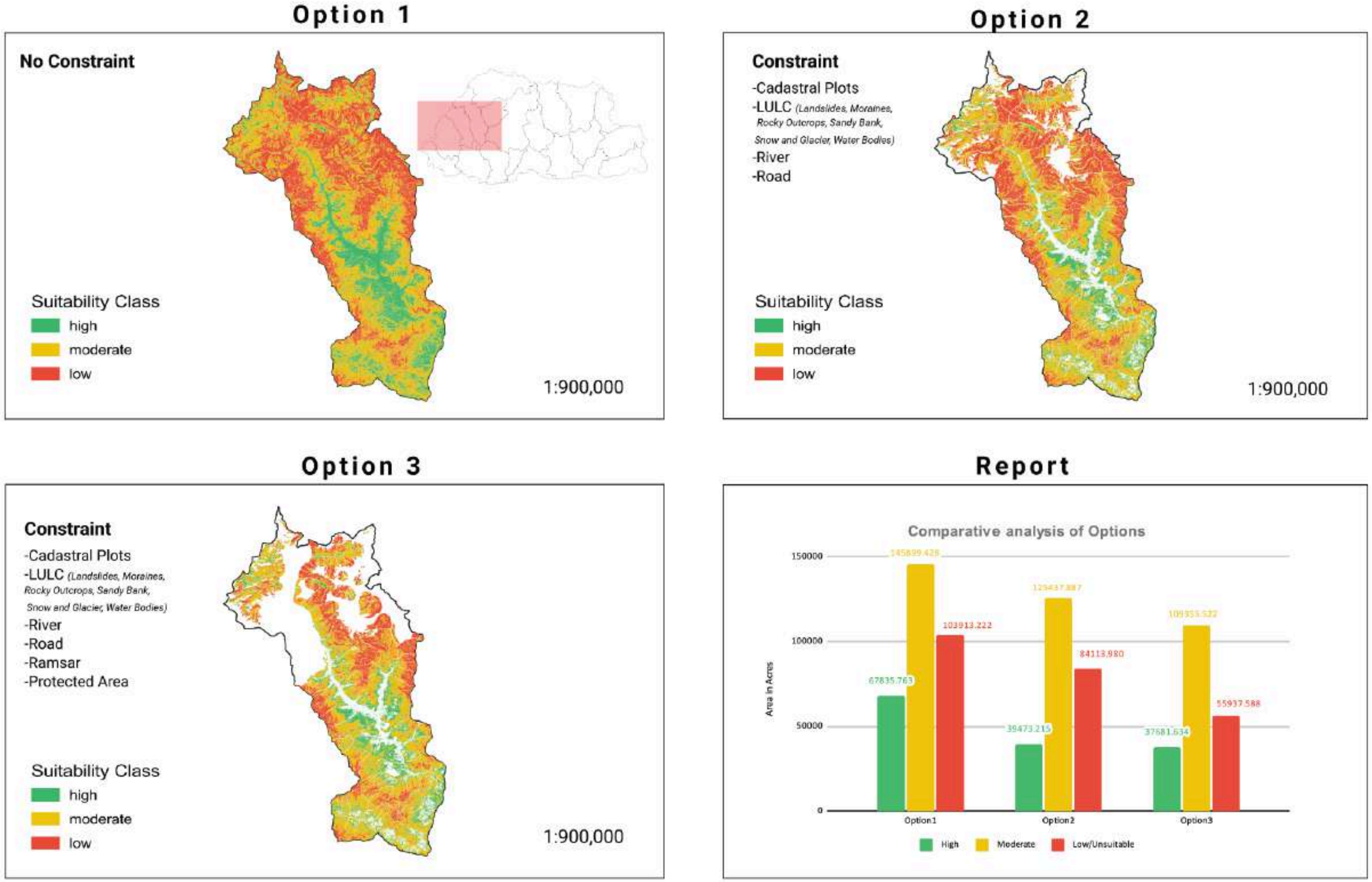


Figure Anx-II-9: Alienability Status of Paro Dzongkhag

Pemagatshel

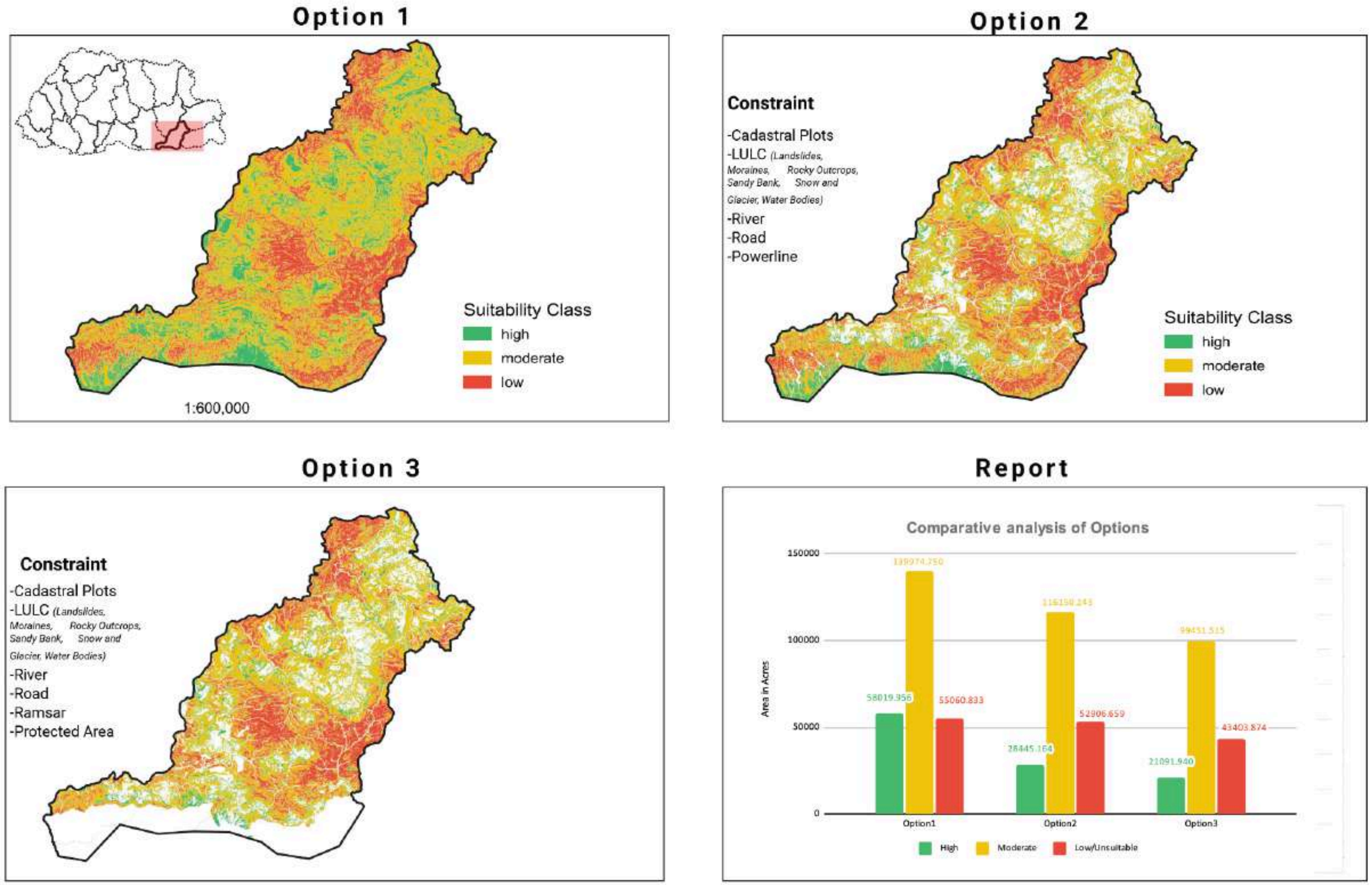


Figure Anx-II-10: Alienability Status of Pemagatshel Dzongkhag

Punakha

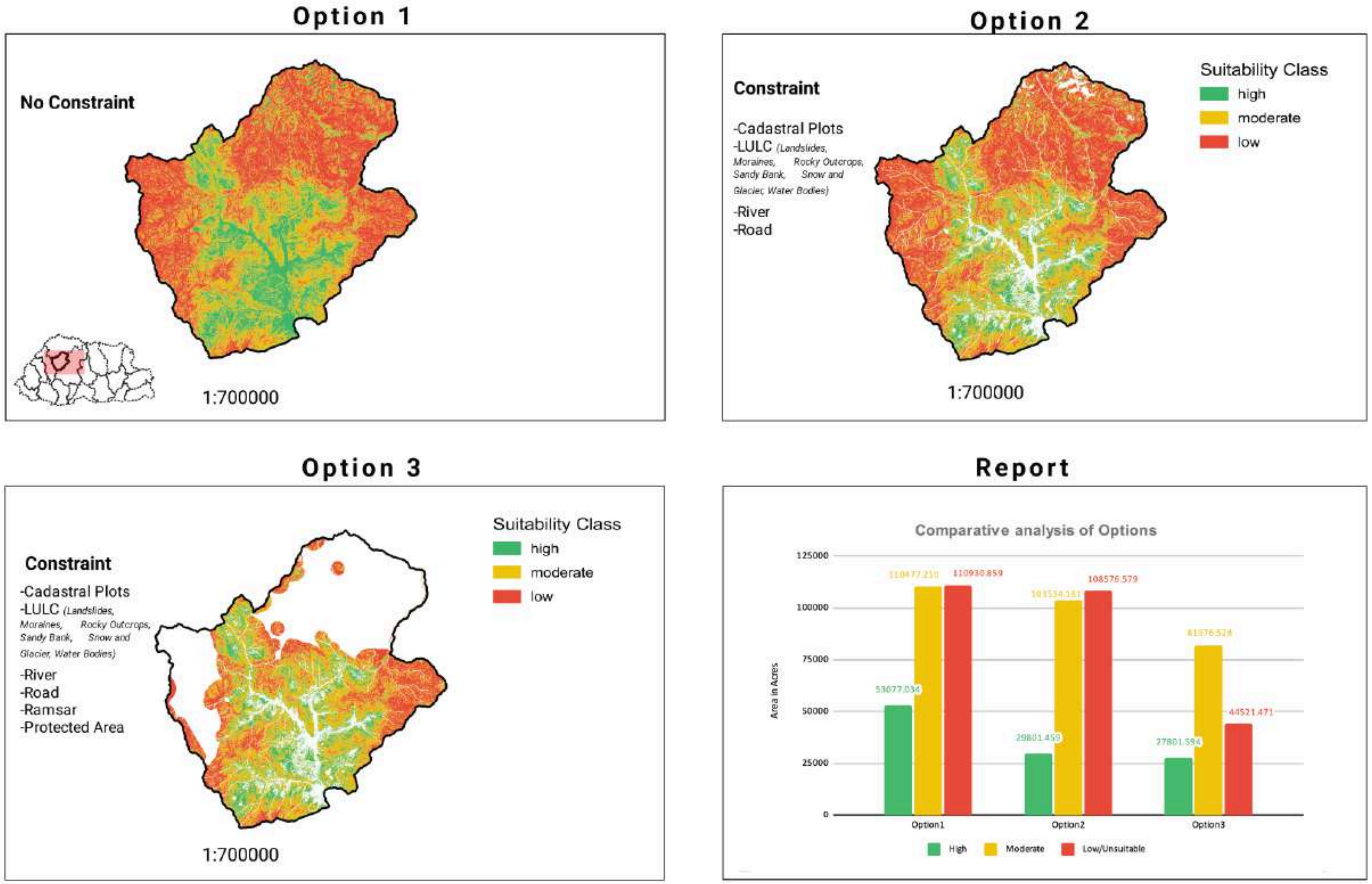


Figure Anx-II-11: Alienability Status of Punakha Dzongkhag

Samdrup Jongkhar

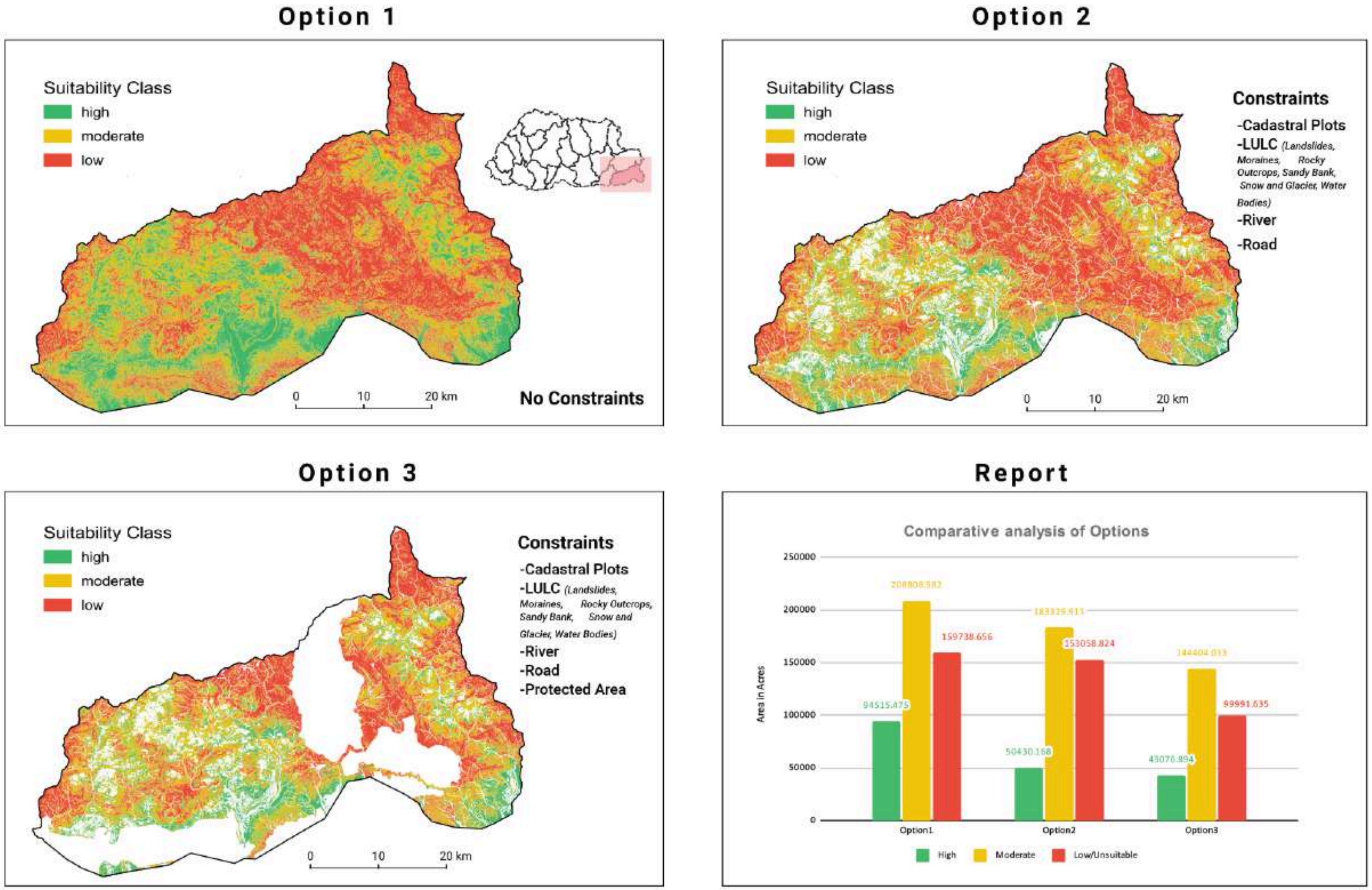


Figure Anx-II-12: Alienability Status of Samdrup Jongkhar Dzongkhag

Samtse

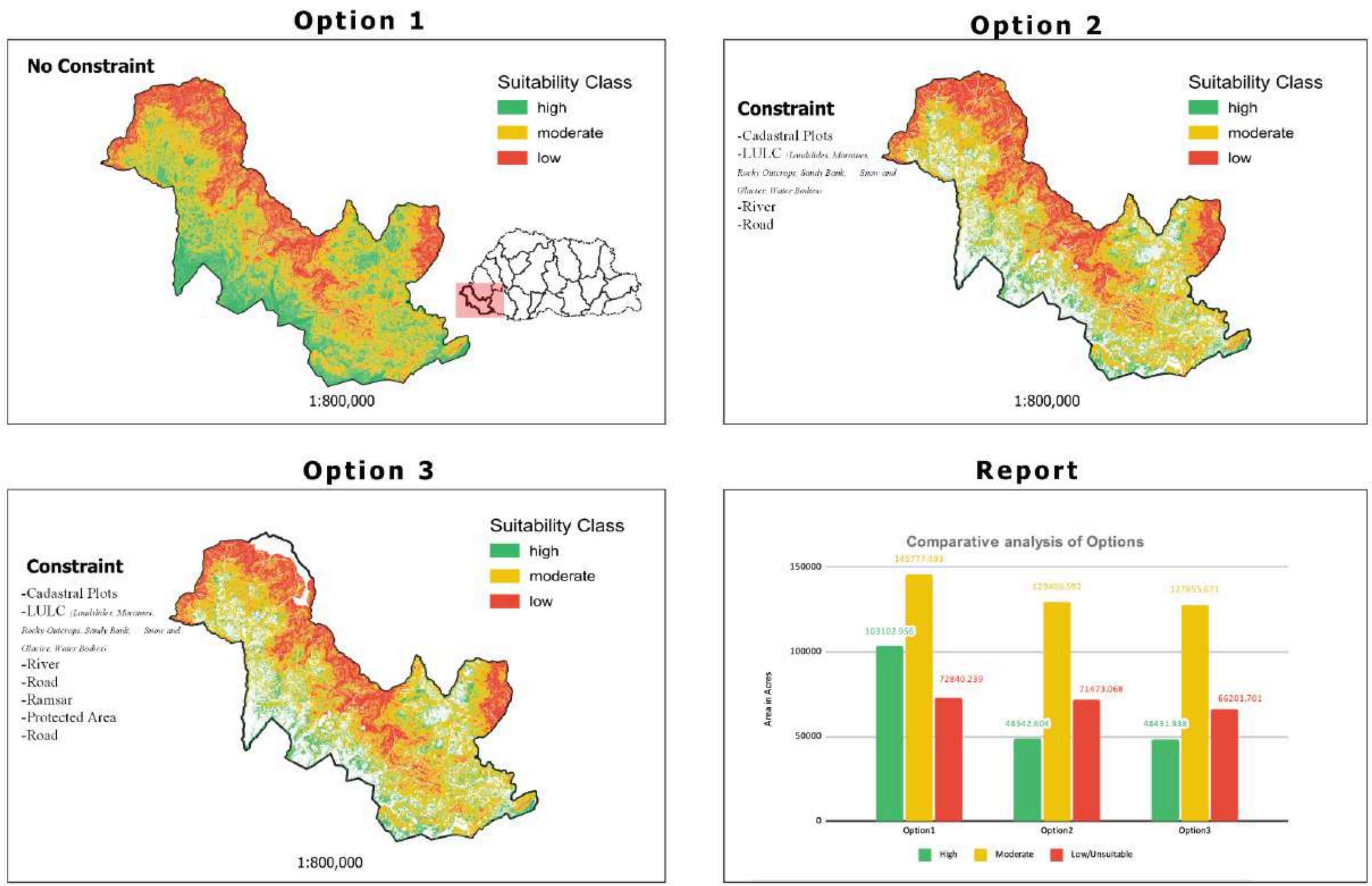


Figure Anx-II-13: Alienability Status of Samtse Dzongkhag

Sarpang

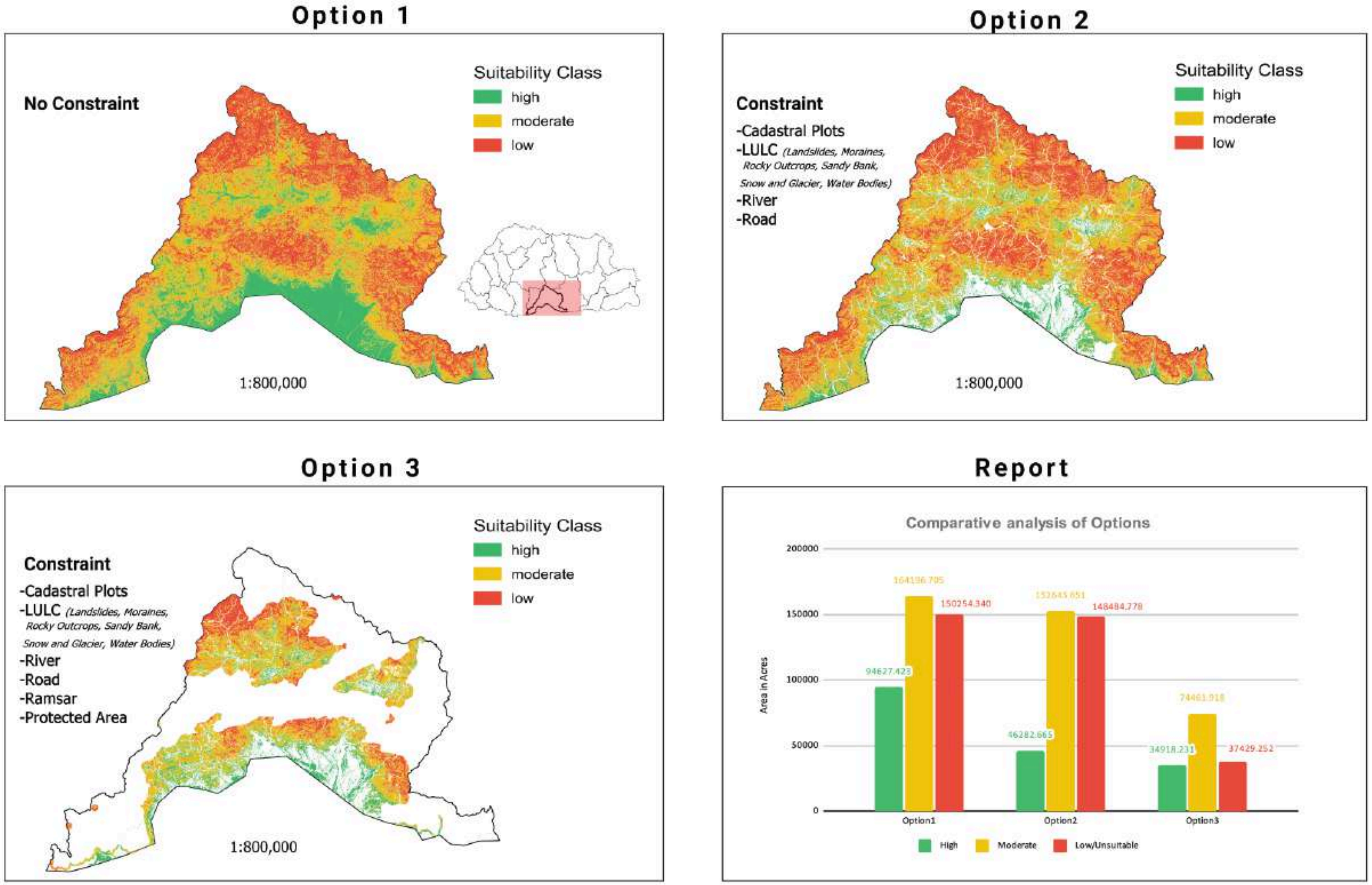


Figure Anx-II-14: Alienability Status of Sarpang Dzongkhag

Tashigang

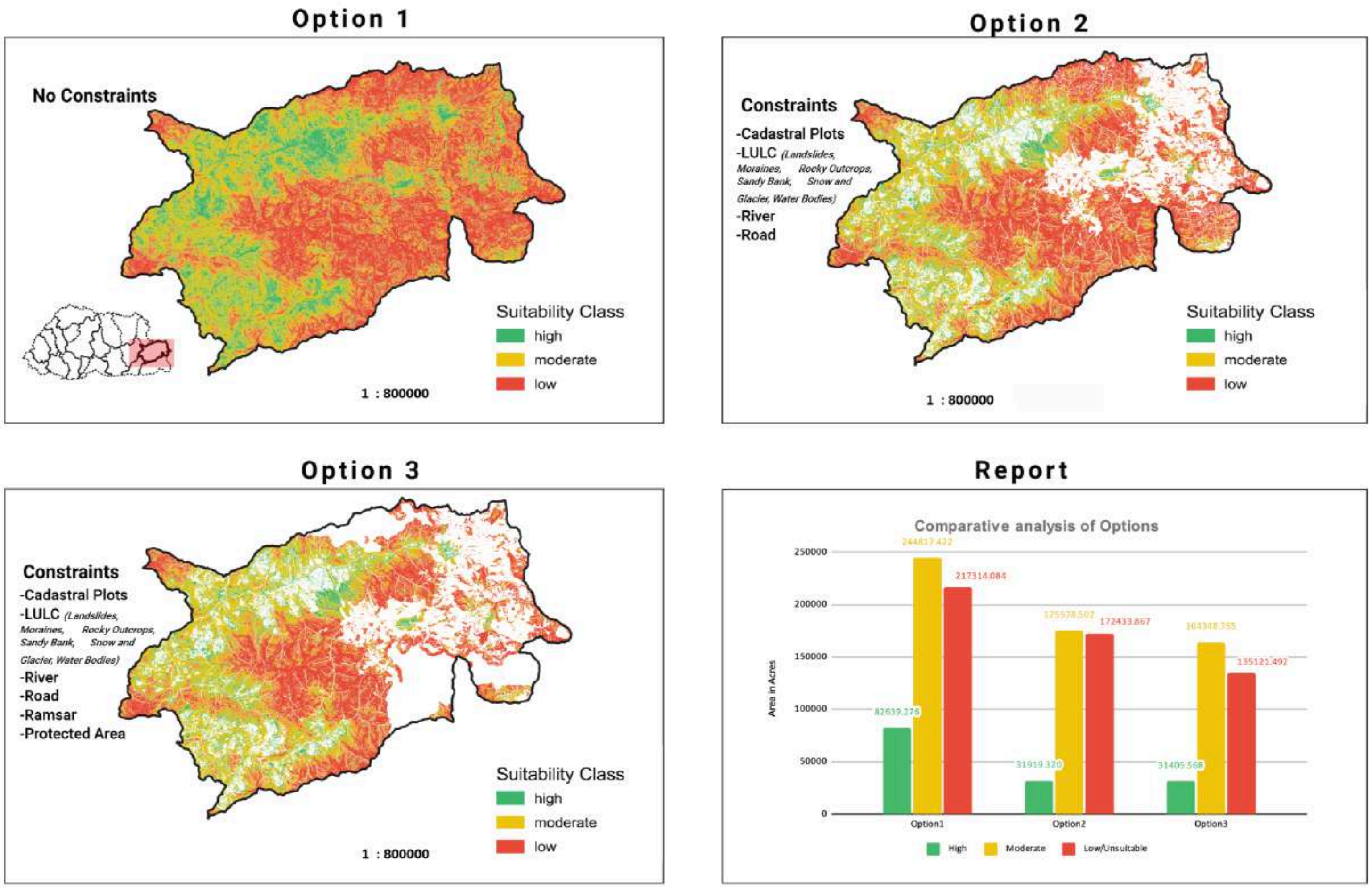
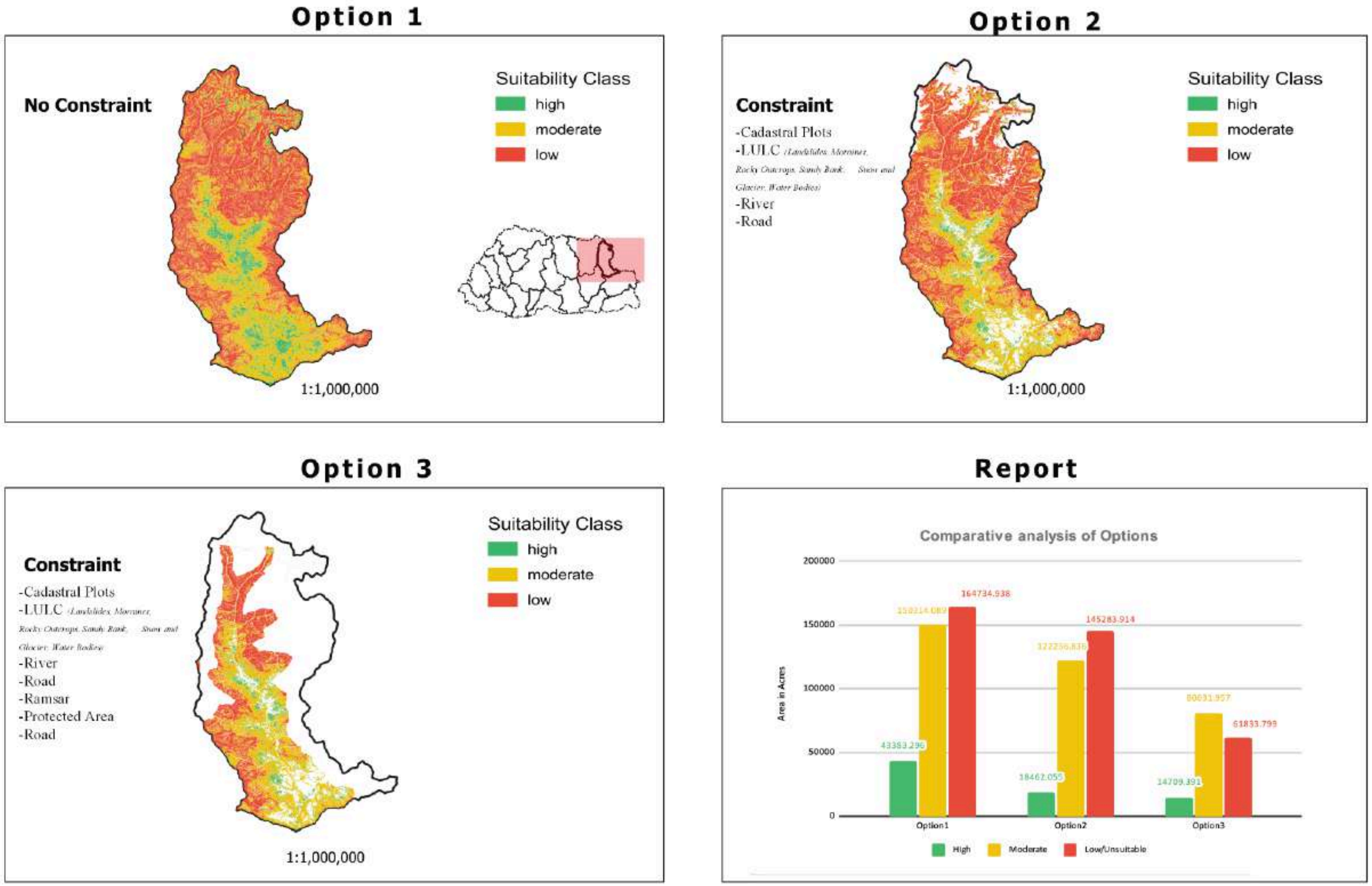


Figure Anx-II-15: Alienability Status of Tashigang Dzongkhag

Tashi Yangtse



FigureAnx-II-16: Alienability Status of Tashi Yangtse Dzongkhag

Thimphu

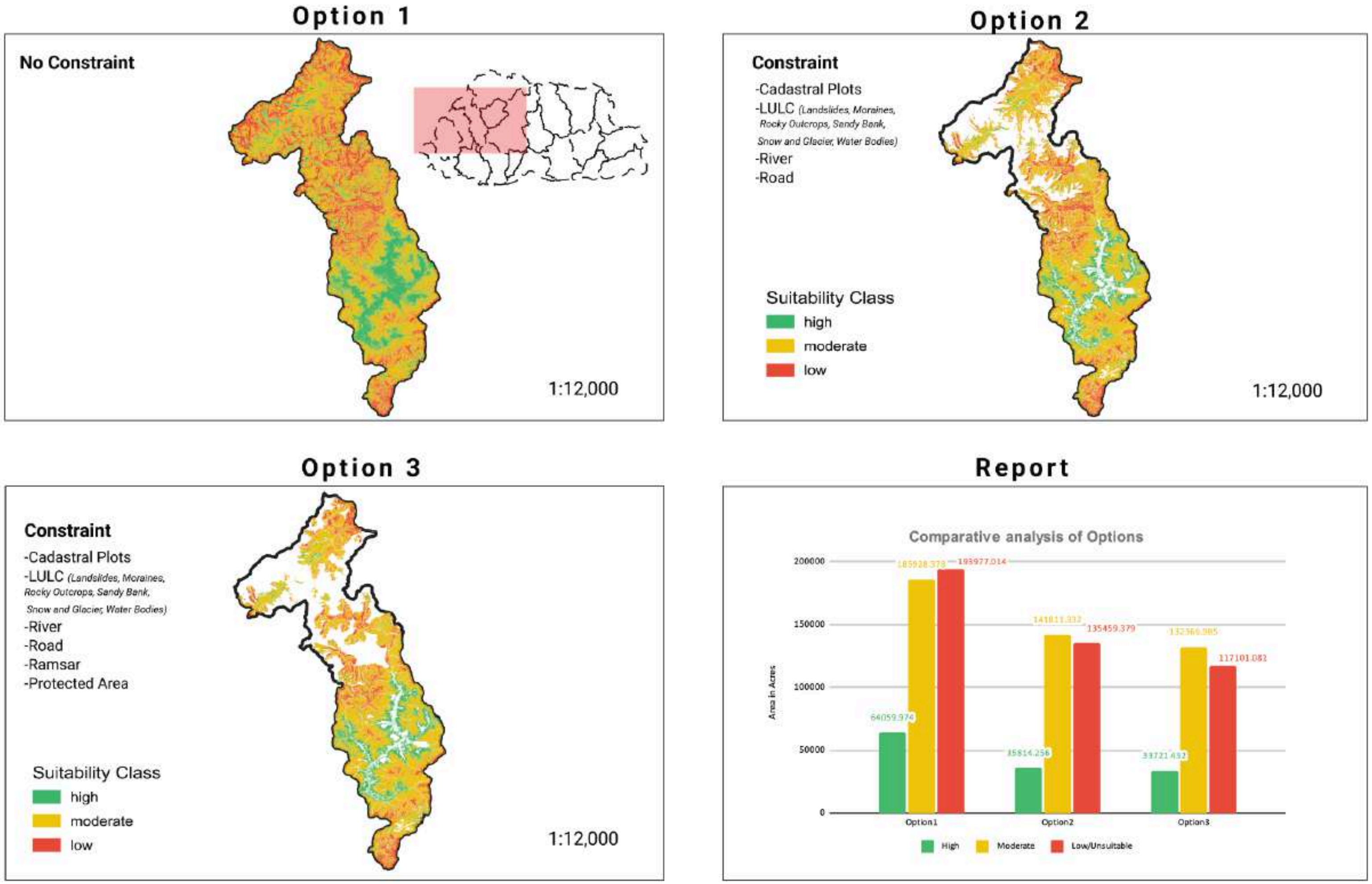


Figure Anx-II-17: Alienability Status of Thimphu Dzongkhag

Trongsa

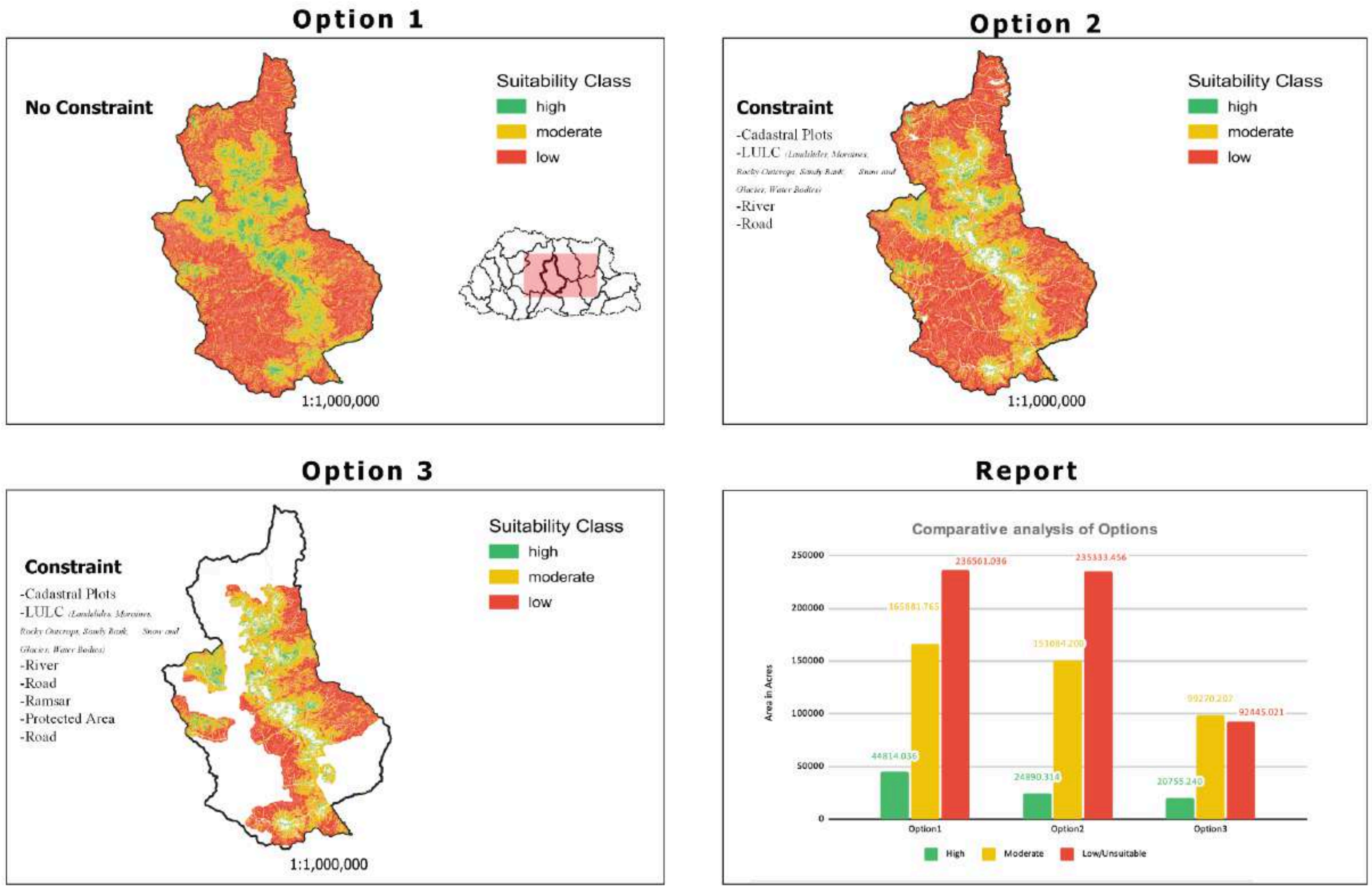


Figure Anx-II-18: Alienability Status of Trongsa Dzongkhag

Tsirang

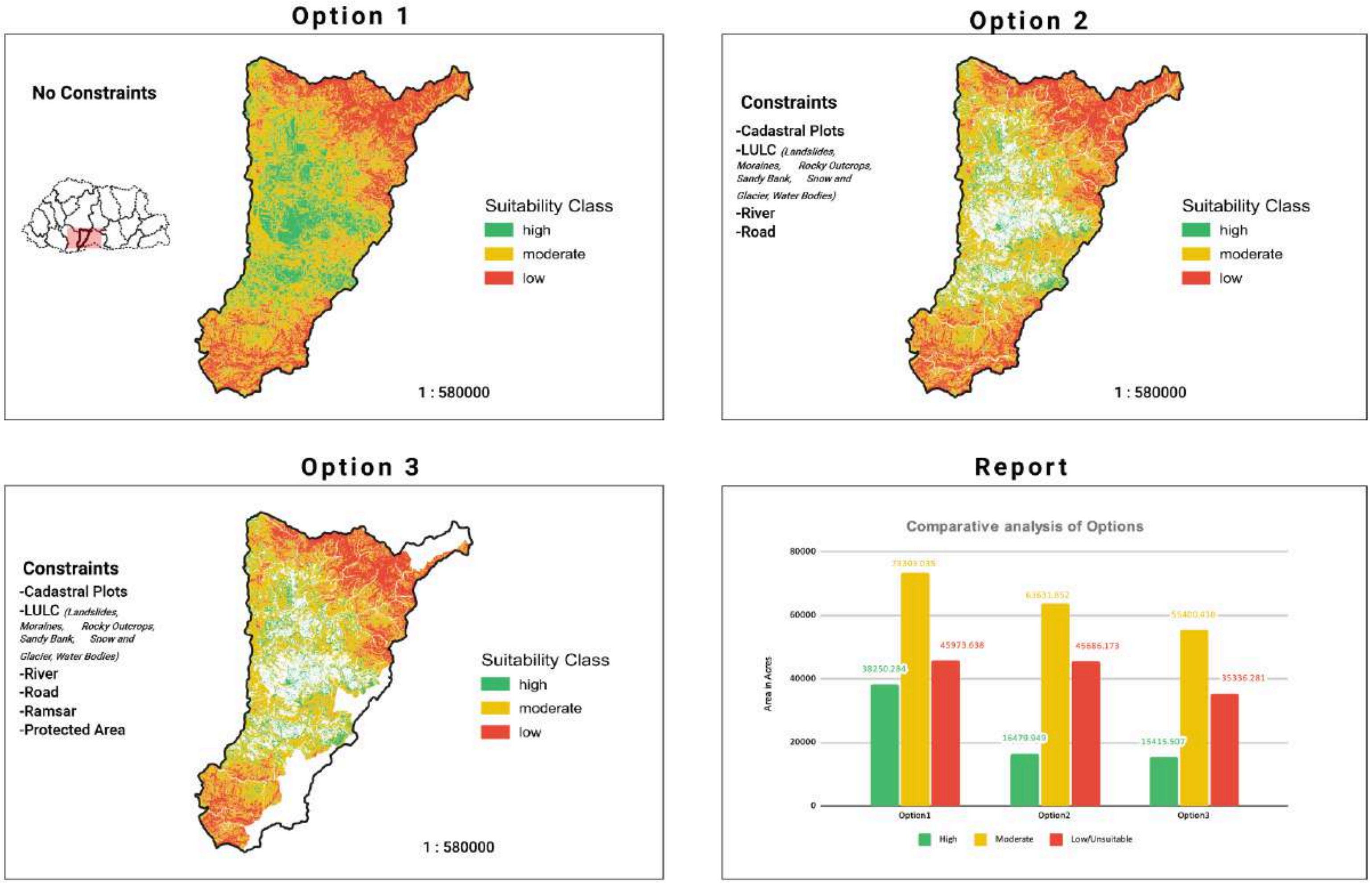


Figure Anx-II-19: Alienability Status of Tsirang Dzongkhag

Wangdue Phodrang

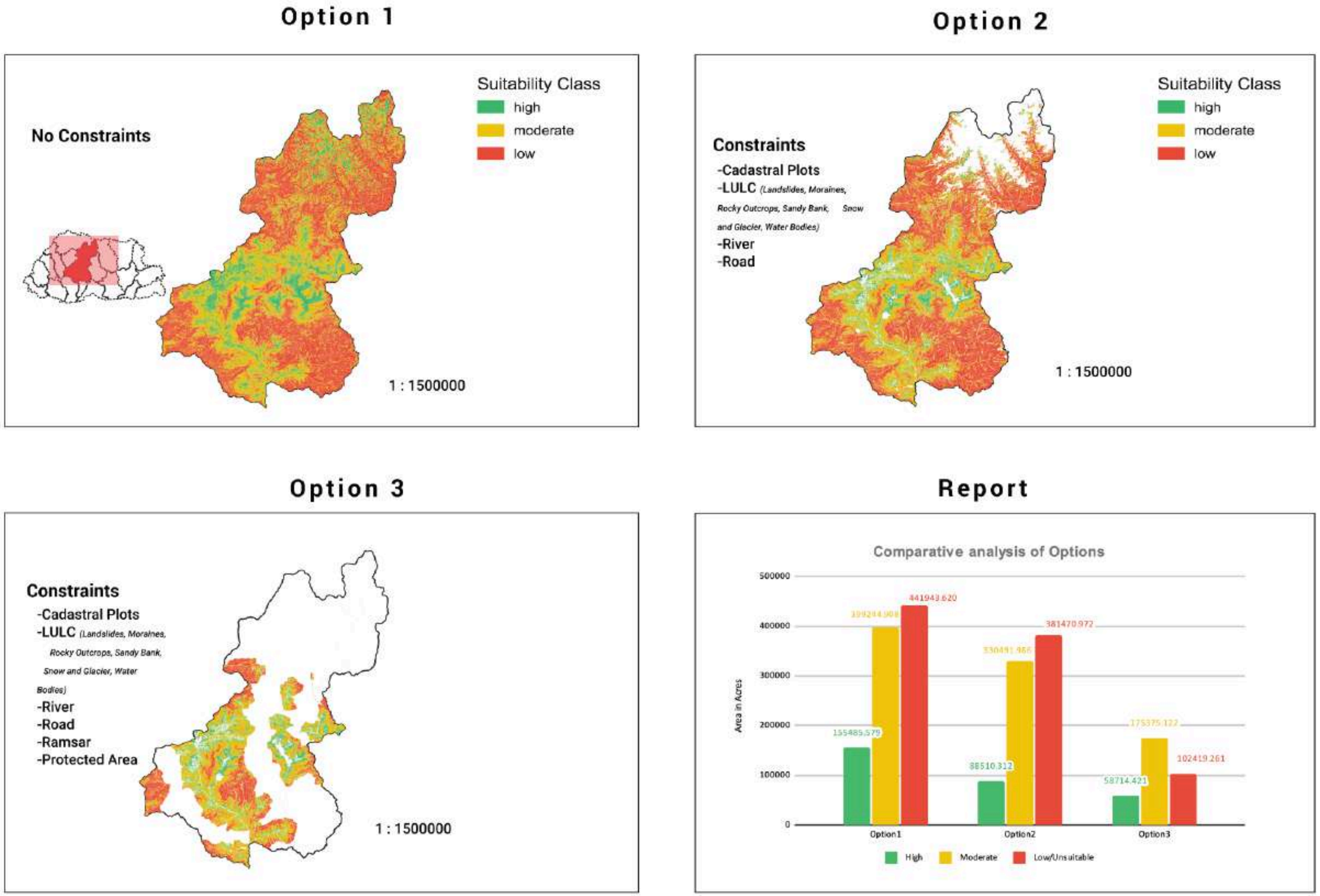


Figure Anx-II-20: Alienability Status of Wangdue Phodrang Dzongkhag

Zhemgang

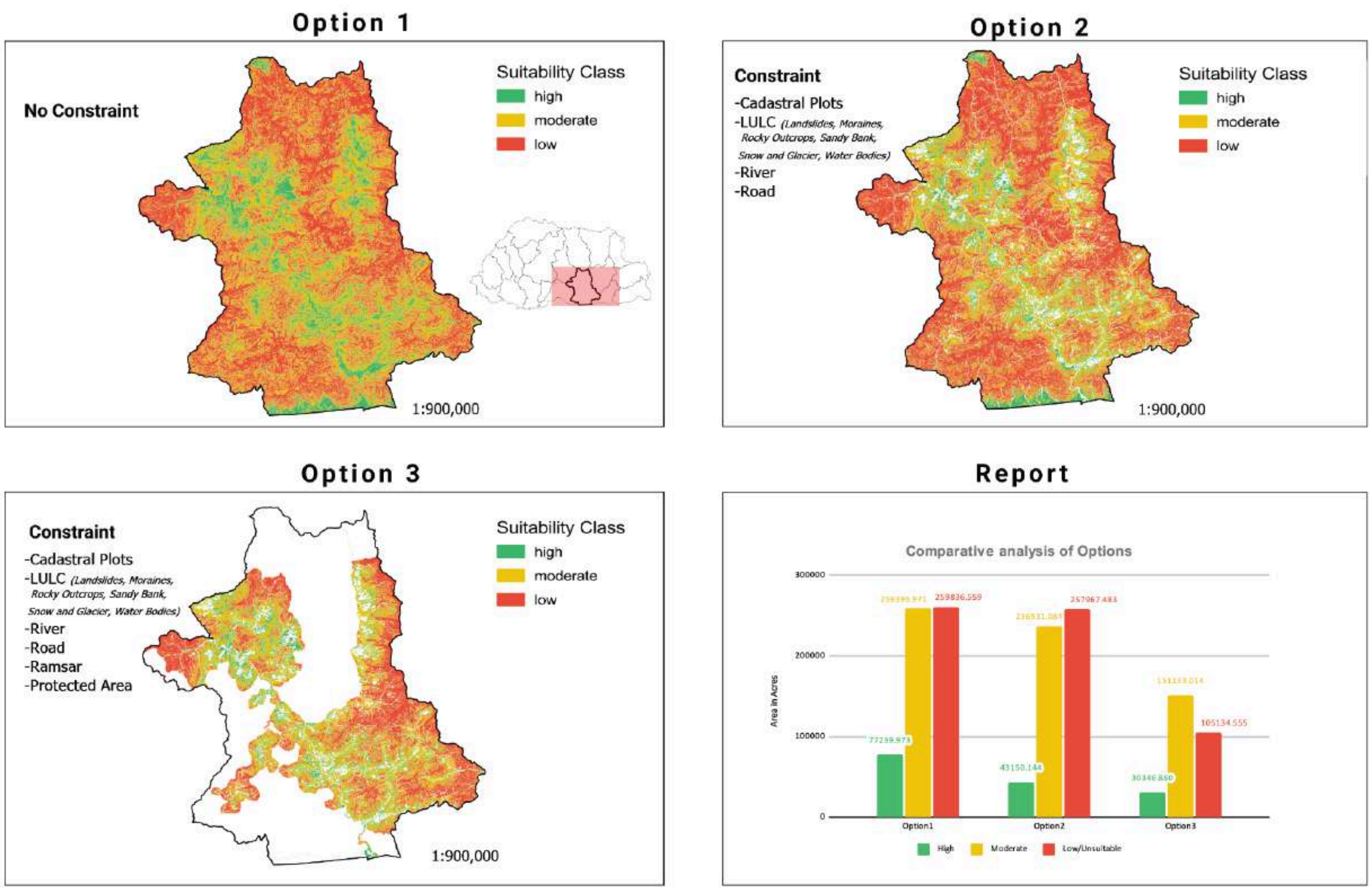


Figure Anx-II-21: Alienability Status of Zhemgang Dzongkhag

APPENDIX-III: MCDA Sensitivity Analysis (±10% Weight Perturbation) and Stability Assessment

This appendix presents the sensitivity analysis of the AHP-based MCDA model using ±10% perturbation of individual factor weights. The objective is to assess the stability and robustness of the spatial suitability outputs under controlled variations in input parameters.

1. Slope Sensitivity

Increment #	Proportion change	slope	settlement	water	road	tcc	dem	Variance	Avg cell difference
1	-0.1	0.44	0.21	0.14	0.09	0.04	0.03	8572.73	57.44
2	0.1	0.53	0.21	0.14	0.09	0.04	0.03	6921.18	49.24
Average					7746.95	53.34			

2. Settlement Sensitivity

Increment #	Proportion change	slope	settlement	water	road	tcc	dem	Variance	Avg cell difference
1	-0.1	0.49	0.19	0.14	0.09	0.04	0.03	13282.27	75.52
2	0.1	0.49	0.23	0.14	0.09	0.04	0.03	7352.54	50.97
Average					10317.4	63.25			

3. Water Sensitivity

Increment #	Proportion change	slope	settlement	water	road	tcc	dem	Variance	Avg cell difference
1	-0.1	0.49	0.21	0.12	0.09	0.04	0.03	4568.61	33.53
2	0.1	0.49	0.21	0.15	0.09	0.04	0.03	9257.63	58.12
Average					6913.12	45.83			

4. Road Sensitivity

Increment #	Proportion change	slope	settlement	water	road	tcc	dem	Variance	Avg cell difference
1	-0.1	0.49	0.21	0.14	0.08	0.04	0.03	5817.02	42.53
2	0.1	0.49	0.21	0.14	0.1	0.04	0.03	2871.95	29.64
Average					4344.49	36.09			

5. Tree Crown Cover (TCC) Sensitivity

Increment #	Proportion change	slope	settlement	water	road	tcc	dem	Variance	Avg cell difference
1	-0.1	0.49	0.21	0.14	0.09	0.04	0.03	6904.22	51.8
2	0.1	0.49	0.21	0.14	0.09	0.05	0.03	9321.81	62.77
Average					8113.01	57.29			

6. Digital Elevation Model (DEM) Sensitivity

Increment #	Proportion change	slope	settlement	water	road	tcc	dem	Varianc e	Avg cell difference
1	-0.1	0.49	0.21	0.14	0.09	0.04	0.03	6373.35	48.29
2	0.1	0.49	0.21	0.14	0.09	0.04	0.03	7267.16	55.5
Average					6820.25	51.9			

Sensitivity Ranking and Overall Model Stability Summary

Rank	Factor	Variance	Average difference	Sensitivity based on variance	Sensitivity based on avg diff
1	Proximity to Settlements	10317.4	63.25	0.233	0.206
2	Tree Canopy Cover (TCC)	8113.01	57.29	0.183	0.186
3	Slope	7746.95	53.34	0.175	0.173
4	Proximity to Water	6913.12	45.83	0.156	0.149
5	Elevation (DEM)	6820.25	51.9	0.154	0.169
6	Proximity to Roads	4344.49	36.09	0.098	0.117
Total		44255.23	307.68	1	1